Admin Platform - Technical Specification KickstartSkills

1.Super Admin Authentication & System Control

1. Super Admin Access: Highest level system access with comprehensive controls

Concept: This is the core account with ultimate power. Think of it as the "root" or "administrator" account for the entire platform.

- **Super Admin Username:** (Text Input) Unique identifier for the Super Admin account (e.g., `superadmin`, `global_admin`).
- **Super Admin Password:** (Password Input) Strong, complex password for initial setup and subsequent changes.
- **Confirm Super Admin Password:** (Password Input) To ensure the password was typed correctly.
- **Super Admin Email Address:** (Email Input) For account recovery and critical notifications.
- **Associated Mobile Number (Optional but Recommended):** (Phone Number Input) For multi-factor authentication (MFA) or emergency contact.
- **Initial Setup Prompt:** (Checkbox/Toggle) "Require Super Admin to change password on first login." (Yes/No)

2. Multi-factor Authentication (MFA): Advanced security with biometric and token support

Concept: Adds extra layers of security beyond just a password. Even if someone steals the password, they still need a second "factor" (like a code from a phone or a fingerprint) to log in.

- **MFA Method Selection:** (Radio Buttons/Dropdown)
  - "Authenticator App (e.g., Google Authenticator, Authy)"
  - "SMS Code to Mobile Number"
  - "Email Code"
  - "Biometric (Fingerprint/Face ID - *Requires device support*)"
  - "Security Key (e.g., YubiKey)"
- **For Authenticator App:**
  - **(Display QR Code/Secret Key):** System-generated output for the admin to scan/input into their app.
- **For SMS Code:**
  - **Mobile Number for SMS:** (Phone Number Input - if not already captured in profile).
- **For Email Code:**
  - **Email Address for Code:** (Email Input - if not already captured in profile).
- **MFA Enforcement Policy:** (Dropdown)

    - "Required for all Super Admins"
    - "Required for all Admins"
    - "Optional"
- **MFA Grace Period (Optional):** (Number Input - e.g., "7 days") Allow users to bypass MFA for a certain period after initial setup.

3. Admin Role Management: Create and manage different admin role hierarchies

Concept: Define different types of administrators (e.g., "Content Admin," "User Manager," "Support Agent") and assign specific permissions to each role. This allows for a structured approach to access control.

- **Role Name:** (Text Input) Unique name for the role (e.g., "Content Editor," "Financial Reviewer").
- **Role Description:** (Text Area) A brief explanation of what this role is for.
- **Base Permissions (Initial Template Selection):** (Dropdown/Checkbox List)
    - "No Permissions (Start from scratch)"
    - "View-Only Access"
    - "Default Admin Template"
    - "Custom Template"
- **Assign Users to Role:** (Multi-select Dropdown/Searchable List) Select existing admin users to assign to this role.
- **Role Status:** (Checkbox/Toggle) "Active/Inactive"

4. System-wide Permissions: Granular permission control across all platform modules

Concept: This is where you specify *exactly* what each role (and by extension, each admin) can *do* within the system. This goes hand-in-hand with "Admin Role Management."

- **Module/Feature Selection:** (Tree View/Categorized List)
    - Users Module:
        - `Can View Users`: (Checkbox)
        - `Can Edit User Profiles`: (Checkbox)
        - `Can Delete Users`: (Checkbox)
        - `Can Reset User Passwords`: (Checkbox)
    - Content Module:
        - `Can Create Posts`: (Checkbox)
        - `Can Publish Posts`: (Checkbox)
        - `Can Edit Own Posts`: (Checkbox)
        - `Can Edit All Posts`: (Checkbox)
    - Settings Module:
        - `Can View General Settings`: (Checkbox)
        - `Can Modify System Configuration`: (Checkbox)

    - (Repeat for all modules: Payments, Reports, Analytics, etc.)
- **Permission Level (if applicable):** (Radio Buttons/Dropdown for more complex permissions)
    - "Read-Only"
    - "Create/Update"
    - "Full Control"
    - "None"
- **Save Permissions Button:** (Action Button)

5. Security Protocols: Advanced security settings and monitoring

Concept: Configuring system-level security rules and keeping an eye on potential threats.

- **Password Policy:**
    - `Minimum Password Length`: (Number Input)
    - `Require Uppercase Letters`: (Checkbox)
    - `Require Lowercase Letters`: (Checkbox)
    - `Require Numbers`: (Checkbox)
    - `Require Special Characters`: (Checkbox)
    - `Password Expiration (Days)`: (Number Input - 0 for never expires)
    - `Prevent Password Reuse (Number of previous passwords)`: (Number Input)
- **Session Management:**
    - `Session Timeout (Minutes)`: (Number Input - e.g., 30, 60)
    - `Idle Session Logout`: (Checkbox)
- **IP Whitelisting/Blacklisting:**
    - `Allowed IP Addresses (Comma-separated)`: (Text Area)
    - `Blocked IP Addresses (Comma-separated)`: (Text Area)
- **Brute Force Protection:**
    - `Max Login Attempts before Lockout`: (Number Input)
    - `Lockout Duration (Minutes)`: (Number Input)
- **SSL/TLS Certificate Management:** (Upload File Input for certificate files, potentially hostname input).
- **Content Security Policy (CSP) Settings:** (Text Area for CSP rules).
- **Email for Security Alerts:** (Email Input) Where critical security notifications are sent.
- **SMS for Critical Security Alerts:** (Phone Number Input)

6. Admin Activity Monitoring: Track all admin activities and system changes

Concept: A detailed log of who did what, when, and where. Essential for auditing, troubleshooting, and security investigations.

- **Activity Log Filters:**
    - `Admin User`: (Dropdown/Searchable List of Admins)

    - `Activity Type`: (Dropdown - e.g., "Login," "User Created," "Settings Changed," "Content Edited," "Role Modified")
    - `Date Range (From/To)`: (Date Pickers)
    - `IP Address`: (Text Input)
    - `Module Affected`: (Dropdown - e.g., "Users," "Content," "Settings")
- **Retention Period for Logs:** (Number Input + Dropdown: Days/Weeks/Months/Years) How long activity logs are kept.
- **Export Log Data:** (Button - e.g., "Export to CSV," "Export to JSON")

7. Emergency Access: Emergency access protocols for critical situations

Concept: A fail-safe mechanism to regain Super Admin access if primary methods fail (e.g., lost MFA device, forgotten password, no other Super Admins available). This should be highly secure and rarely used.

- **Emergency Contact Person(s) Name:** (Text Input)
- **Emergency Contact Person(s) Phone Number:** (Phone Number Input)
- **Emergency Contact Person(s) Email:** (Email Input)
- **Emergency Access Token/Key Generation:** (Button to Generate Token) - This would be a unique, one-time use code or key displayed for secure storage (e.g., print and store in a safe).
- **Verification Questions for Emergency Access:** (Text Area for custom questions and answers) - e.g., "What was the name of the first module created?"
- **Required Approvals for Emergency Access:** (Number Input) - e.g., "Requires 2 out of 3 designated approvers."
- **Notification Email for Emergency Access Use:** (Email Input) Alert when emergency access is invoked.

8. Admin Account Recovery: Secure admin account recovery procedures

Concept: A process for individual admins (including Super Admins) to regain access to their own accounts if they forget their password or lose their MFA device, without relying solely on emergency access.

- **Recovery Method Prioritization:** (Draggable List/Dropdowns)
    - "Email Verification" (Send code to registered email)
    - "SMS Verification" (Send code to registered phone)
    - "Security Questions" (Prompt pre-set questions)
    - "Admin Approval" (Requires another admin to approve the recovery)
- **Security Questions Setup (User-facing during profile setup):**
    - `Question 1 Selection`: (Dropdown of common security questions)
    - `Answer 1`: (Text Input)
    - `Question 2 Selection`: (Dropdown)
    - `Answer 2`: (Text Input)
    - (Repeat for 3-5 questions)

- **Email Template for Recovery:** (Rich Text Editor for customizing recovery emails).
- **Recovery Flow Configuration:** (Checkboxes/Toggles)
    - "Allow self-service password reset via email."
    - "Allow self-service MFA reset via security questions."
    - "Require admin approval for MFA reset."
- **Notification Email for Account Recovery Attempt:** (Email Input) To alert admins of recovery attempts on their accounts.

2.Comprehensive System Dashboard

1. System Health Monitoring

- **What it shows:** If the website and its parts are working smoothly.
- **"Input" (Configuration Needed):**
    - **Server IP Addresses/URLs:** Where are our servers located on the network? (e.g., `192.168.1.1`, `api.kickstartskills.com`)
    - **Monitoring Agent Credentials:** How can our monitoring tools securely connect to these servers? (e.g., API keys, usernames/passwords for monitoring agents)
    - **Alert Thresholds:** At what point should the system tell us there's a problem? (e.g., "Alert if CPU usage is over 80% for 5 minutes," "Alert if server memory is below 10%")
    - **Service Port Numbers:** Which specific services (like the website, database, etc.) on the server should we monitor? (e.g., Port 80 for web, Port 3306 for MySQL)

2. User Activity Overview

- **What it shows:** What users are doing right now on the platform.
- **"Input" (Configuration Needed):**
    - **Log File Paths:** Where does the system record what users do? (e.g., `/var/log/nginx/access.log`, database tables storing user actions)
    - **User Action Definitions:** What specific actions do we want to track? (e.g., "Course Enrollment," "Internship Application," "Lesson Completed," "Login Success")
    - **Real-time Stream Endpoints:** If we're using a real-time data system, where does the user activity data flow into? (e.g., a Kafka topic, a WebSocket endpoint)

3. **Platform Statistics**

- **What it shows:** Numbers about our users, courses, and internships.
- **"Input" (Configuration Needed):**

- **Database Connection Strings:** How do we connect to the database where user, course, and internship information is stored? (e.g., `mysql://user:password@host:port/database`)
- **Table/Collection Names:** Which specific tables or collections in the database contain this data? (e.g., `users`, `courses`, `internships`)
- **Filtering Criteria:** Do we want to exclude certain types of users (e.g., admin accounts from general statistics)?

4. **Performance Metrics**

- **What it shows:** How fast and efficient our system is.
- **"Input" (Configuration Needed):**
  - **Application Performance Monitoring (APM) Tool Integration:** If we're using tools like New Relic, Datadog, where are their API keys or integration points?
  - **Endpoint URLs to Monitor:** Which specific website pages or API services should we measure the load time for? (e.g., `/login`, `/dashboard`, `/api/courses`)
  - **Response Time Thresholds:** What's considered a "good" or "bad" response time? (e.g., "Alert if login takes longer than 2 seconds")

5. **Error Monitoring**

- **What it shows:** When and where errors are happening.
- **"Input" (Configuration Needed):**
  - **Error Logging Paths/Services:** Where do our applications send their error messages? (e.g., `sentry.io` API key, ELK stack endpoint, specific log files)
  - **Error Severity Levels:** Which types of errors do we want to display (e.g., Critical, Warning, Info)?
  - **Exclusion Rules:** Are there certain "expected" errors we want to ignore?

6. **Resource Utilization**

- **What it shows:** How much server power and database space we're using.
- **"Input" (Configuration Needed):**
  - **Cloud Provider APIs (if applicable):** If on AWS, Azure, GCP, connection details to pull resource usage data.
  - **Server Agent Configuration:** For on-premise servers, settings for agents that report CPU, RAM, disk usage.
  - **Database Connection Details:** To query database size, active connections, etc. (e.g., for MySQL: `SHOW STATUS`, `SHOW VARIABLES`)

7. **Traffic Analytics**

- **What it shows:** How many people are visiting and how they move around the platform.
- **"Input" (Configuration Needed):**
  - **Web Server Log Locations:** Where are the records of every visitor to our website? (e.g., Apache access logs, Nginx access logs)
  - **Analytics Tool Integration:** If using Google Analytics, Matomo, etc., API keys or tracking IDs.
  - **Traffic Source Definitions:** How do we categorize where visitors come from (e.g., "Organic Search," "Social Media," "Direct")?

8. **Revenue Dashboard**

- **What it shows:** How much money we're making in real-time.
- **"Input" (Configuration Needed):**
  - **Payment Gateway API Keys:** How do we securely connect to payment providers (e.g., Stripe, Razorpay, PayPal) to get transaction data?
  - **Order/Transaction Database Tables:** Where do we store successful payment records in our own database?
  - **Product/Service Pricing Rules:** How do we know how much each course or internship costs if not directly from transaction data? (e.g., a lookup table for course prices)
  - **Currency Conversion Rates (if applicable):** If dealing with multiple currencies, where do we get the latest exchange rates?

3. User Management & Role Administration

1. User Creation / Editing (Multi-role User Management)

This is for adding new users or modifying existing user details.

Core User Information:

- **Full Name:** (Text Field) e.g., "John Doe"
- **Email Address:** (Text Field) e.g., "john.doe@example.com" (Must be unique)
- **Phone Number:** (Text Field) e.g., "+919876543210" (Optional, but good for contact)
- **Password:** (Password Field) (Required for new users, confirm password field also needed)
- **Confirm Password:** (Password Field) (Matches the above)
- **User Type/Role:** (Dropdown/Radio Buttons - *Crucial for Multi-role management*)
  - Student
  - Institution
  - Mentor
  - HR

    - Admin Staff
    - *... (Any other custom roles)*

Role-Specific Information (Conditional based on "User Type/Role"):

- **If User Type is "Student":**
    - **Student ID/Roll Number:** (Text Field) e.g., "KS2023001" (If applicable)
    - **Institution Affiliation:** (Dropdown/Search Field) Link to an existing institution.
    - **Course/Program:** (Text Field/Dropdown) e.g., "B.Tech Computer Science"
    - **Enrollment Date:** (Date Picker)

- **If User Type is "Institution":**
    - **Institution Name:** (Text Field) e.g., "ABC University"
    - **Institution Type:** (Dropdown) e.g., "University", "College", "Training Center"
    - **Website:** (Text Field) e.g., "www.abcuni.edu"
    - **Contact Person Name:** (Text Field)
    - **Contact Person Email:** (Text Field)

- **If User Type is "Mentor":**
    - **Specialization/Expertise:** (Multi-select Checkboxes/Text Area) e.g., "Web Development, Data Science, Career Counseling"
    - **Company/Organization (Optional):** (Text Field)
    - **Experience Level:** (Dropdown) e.g., "Junior", "Mid-level", "Senior"

- **If User Type is "HR":**
    - **Company Name:** (Text Field)
    - **Department (Optional):** (Text Field)

- **If User Type is "Admin Staff":**
    - **Admin Department (Optional):** (Text Field) e.g., "IT", "Operations"

2. Bulk User Operations (Import)

This is for uploading many users at once.

- **File Upload:** (File Input Field) Accepts specific formats like:
  - **.CSV file:** (Comma Separated Values) - Most common for bulk uploads.
  - **.XLSX file:** (Excel Spreadsheet) - Often preferred for larger datasets.
- **Template Download Link:** (Button/Link) To provide users with a pre-formatted template for their bulk data, showing required columns (e.g., Name, Email, Role, etc.). This ensures correct data entry.

3. User Verification

These fields are typically used by an administrator to manage a user's lifecycle.

- **User ID/Search Filter:** (Text Field) To find a specific user.
- **Verification Status:** (Dropdown/Radio Buttons)
  - Pending Verification
  - Verified
  - Rejected

4. Account Status Management

- **Account Status:** (Dropdown/Radio Buttons)
  - Active
  - Deactivated
  - Suspended
  - Archived (Optional)
- **Reason for Suspension/Deactivation (Optional):** (Text Area) For admin notes.

5. Role Assignment & Permission Management

This section defines what a user can *do* on the platform.

- **User Selection:** (Search Field/Dropdown) To choose which user's roles/permissions to modify.
- **Available Roles:** (Multi-select Checkboxes/List with "Add/Remove" buttons)
  - Student
  - Institution Admin
  - Mentor
  - HR Manager

    - Super Admin
    - *... (Custom roles)*
- **Specific Permissions (Granular Control - Advanced):** (Multi-select Checkboxes/Toggle Switches)
    - View Analytics
    - Edit User Profiles
    - Approve Content
    - Manage Courses
    - Access Financial Reports
    - *... (Specific actions within the platform)*
    - *(Often, permissions are tied to roles, so directly editing permissions for a single user might be less common than assigning them a role.)*

6. User Analytics

These are typically display/action buttons rather than input fields for *creating* data, but they involve selecting parameters for output.

User Analytics (Filtering & Reporting):

- **Date Range:** (Date Picker - From/To) To view analytics for a specific period.
- **User Type Filter:** (Dropdown/Multi-select) To filter analytics by role.
- **Engagement Metric:** (Dropdown) e.g., "Logins", "Courses Completed", "Messages Sent".

7.Account Security (Monitoring & Management):

- **User Search:** (Text Field) To find a user for security review.
- **Login History View:** (Table/Display) Shows IP addresses, login times. (No input, but part of the display).
- **Force Password Reset:** (Button) For admin to trigger a password reset for a user.
- **Session Management:** (List of active sessions with "End Session" button) For admin to terminate suspicious sessions.

8. Data Export

- **Export Format:** (Dropdown) For data export, usually presented as buttons:
    - CSV
    - Excel (.xlsx)
    - PDF (for reports)

4. Course & Content Management System

1. Course Creation Tools (Creating a New Course)

- **Course Title:** (Text Input)
- **Course Description:** (Text Area)
- **Course Thumbnail/Image:** (File Upload)
- **Course Category:** (Dropdown/Multi-select) - e.g., Programming, Marketing, Design.
- **Course Instructor(s):** (Multi-select from existing Mentors/Instructors)
- **Course Price/Access:** (Number Input/Radio Button) Free, Paid, Subscription Required.
- **Course Duration (Optional):** (Text Input) - e.g., "10 hours," "4 weeks."
- **Learning Outcomes:** (Text Area/Repeater Field) - Each outcome as a separate input.
- **Course Status:** (Dropdown) Draft, Published, Archived.

2. Video Management (Uploading a New Video)

- **Video File:** (File Upload) - Accepts various video formats.
- **Video Title:** (Text Input)
- **Video Description:** (Text Area)
- **Associated Course/Lesson:** (Dropdown/Search Input) - to link the video to specific content.
- **Visibility:** (Radio Button) Public, Private, Course-specific.
- **Subtitles/Captions (Optional):** (File Upload) - SRT, VTT files.

3. Content Library (Adding/Managing General Content)

- **Content Type:** (Dropdown) Document (PDF, DOCX), Image (JPG, PNG), Audio (MP3), Other.
- **Content File:** (File Upload)
- **Content Title:** (Text Input)
- **Content Description:** (Text Area)
- **Tags/Keywords:** (Text Input with auto-suggest)
- **Version Notes:** (Text Area) - when updating existing content.

4. Course Curriculum Builder (Adding a New Lesson/Module)

- **(Within a Course Editor Interface)**
- **Module Title:** (Text Input)
- **Module Description:** (Text Area)
- **Lesson Title:** (Text Input)
- **Lesson Type:** (Dropdown) Video, Text, Quiz, Assignment, External Link.
- **Associated Content:** (Search/Select) - Link to existing videos, documents, or create new.
- **Order/Sequence:** (Drag-and-drop handles for reordering modules and lessons)

5. Assessment Creation (Creating a New Quiz/Assessment)

- **Assessment Title:** (Text Input)

- **Assessment Description:** (Text Area)
- **Assessment Type:** (Dropdown) Multiple Choice, True/False, Short Answer, Essay, Coding Challenge.
- **Associated Lesson/Module:** (Dropdown/Search Input)
- **Pass Mark/Score:** (Number Input)
- **Time Limit (Optional):** (Number Input) - in minutes.
- **Question Text:** (Text Area) - (Repeater Field for multiple questions)
- **Answer Options (for MCQs):** (Text Input with Radio Button for correct answer) - (Repeater Field for multiple options)
- **Feedback (for correct/incorrect answers):** (Text Area - Optional)

6. **Content Approval Workflow (Submitting Content for Review)**

- **(This is more of a process than direct input fields, but involves status changes)**
- **Current Status:** (Display Only) Draft, Pending Review, Approved, Rejected.
- **Submit for Review Button:** (Action Button)
- **Reviewer Comments (for Reviewers):** (Text Area)
- **Approve/Reject Button (for Reviewers):** (Action Buttons)

7. Content Analytics: Track Content Performance and Engagement Metrics

This section is primarily about *viewing* data, not *inputting* data.

What you'd be looking to view (not input):

- Number of views/plays
- Completion rates
- Engagement time
- User feedback/ratings
- Assessment scores
- Common drop-off points in videos

8. **Multilingual Support (Adding a New Language Version for Content)**

- **Original Content ID:** (Display Only)
- **Language:** (Dropdown) - e.g., Hindi, Marathi, English, Spanish.
- **Translated Title:** (Text Input)
- **Translated Description:** (Text Area)
- **Translated Content File (if applicable):** (File Upload)

9. Content Delivery Optimization: CDN Integration for Optimal Content Delivery

This is largely a technical setup and configuration, not something a daily administrator would typically input.

What might be configured by a technical administrator (not regular input fields):

- **CDN Provider Details:** (e.g., Cloudflare, Akamai credentials)
- **Domain Configuration:** Linking the CDN to your platform's domain.
- **Caching Rules:** How long content should be cached.

5. Advanced Examination Management

1. Exam Creation Interface: Advanced exam creation with multiple question types

- **What you'll input:**
  - **Exam Title:** (e.g., "Mid-Term Python Programming," "Final Exam - Data Structures")
  - **Exam Description:** (Brief explanation of the exam)
  - **Exam Duration:** (How long students have to complete it, e.g., "60 minutes," "2 hours")
  - **Start Date & Time:** (When the exam becomes available)
  - **End Date & Time:** (When the exam closes)
  - **Total Marks:** (Maximum marks for the exam)
  - **Passing Marks/Percentage:** (What score students need to pass)
  - **Question Type Selection:** (Checkboxes/dropdowns for "Multiple Choice," "True/False," "Short Answer," "Essay," "Code Snippet," etc.)
  - **Instructions for Students:** (Any specific rules or guidelines for taking the exam)
  - **Randomize Question Order:** (Yes/No toggle)
  - **Randomize Option Order (for MCQs):** (Yes/No toggle)
  - **Number of Questions per Section/Type:** (If you want to pull a certain number of questions from different categories)

2. Question Bank Management: Comprehensive question bank with categorization

- **What you'll input:**
  - **Question Text:** (The actual question, e.g., "What is the capital of France?")
  - **Question Type:** (Dropdown: "Multiple Choice," "True/False," "Short Answer," etc. - *This links to the Exam Creation Interface*)
  - **Category/Topic:** (Dropdown/Text Field: "History," "Mathematics - Algebra," "Programming - Python Basics," "Cybersecurity - Network Security")
  - **Difficulty Level:** (Dropdown: "Easy," "Medium," "Hard")
  - **Marks for this Question:** (How many marks this specific question is worth)
  - **Correct Answer(s):**
    - **For Multiple Choice:** List of options, with a checkbox/radio button to select the correct one(s).
    - **For True/False:** Radio button for "True" or "False."

        - **For Short Answer/Essay:** (Optional) Keywords or a sample answer for auto-grading or reviewer guidance.
        - **For Code Snippet:** Test cases, expected output.
    - **Explanation/Feedback (Optional):** (Why an answer is correct/incorrect, shown after results)
    - **Tags:** (Keywords to help find questions, e.g., "Loops," "Variables," "French Revolution")

3. Anti-cheating Measures: Proctoring, browser lockdown, plagiarism detection

- **What you'll input:**
    - **Enable Proctoring:** (Toggle: "Live Proctoring," "Automated AI Proctoring," "No Proctoring")
        - *If Live:* Setup proctor schedule, assign proctors.
        - *If Automated AI:* Configure sensitivity settings (e.g., "Flag suspicious eye movements," "Detect multiple faces").
    - **Enable Browser Lockdown:** (Toggle: Yes/No)
        - *If Yes:* List of allowed/disallowed applications during the exam.
    - **Enable Plagiarism Detection (for Essay/Code):** (Toggle: Yes/No)
        - *If Yes:* Set similarity threshold (e.g., "Flag if over 20% similar to other submissions/online sources").

4. Exam Scheduling: Automated exam scheduling and calendar management

- **What you'll input:**
    - **Select Exam:** (Dropdown list of exams you've created)
    - **Assign to Students/Groups:** (Selection of individual students or pre-defined student groups/batches)
    - **Override Exam Start/End Times (for specific groups):** (If different groups have different schedules)
    - **Reminders:** (Set up automated email/notification reminders for students before the exam)

5. Result Processing: Automated result calculation and grade processing

- **What you'll input:** (Less direct input here, more configuration of how results are processed)
    - **Visibility of Results:** (Toggle: "Immediately after submission," "After manual review," "On a specific date")
    - **Show Correct Answers to Students:** (Toggle: "Yes," "No," "After results are published")
    - **Show Explanation/Feedback to Students:** (Toggle: "Yes," "No")
    - **Export Options:** (Buttons to export results in CSV, PDF, Excel)
    - **Integration with Gradebook:** (Settings to push grades to an external gradebook system if applicable)

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6. Performance Analytics: Detailed exam performance analytics and insights

- **What you'll input:** (Mainly filters and selections to *view* analytics)
  - **Select Exam:** (Dropdown)
  - **Select Group/Student:** (Dropdown)
  - **Date Range:** (To view performance over time)
  - **Filter by Question Category/Difficulty:** (To see performance on specific topics)

7. Exam Security: Advanced security measures for exam integrity

- **What you'll input:** (These are mostly behind-the-scenes, but some configuration)
  - **Access Control:** (Assign roles and permissions to admins/staff for exam management - e.g., "Exam Creator," "Result Reviewer," "Proctor")
  - **IP Restrictions:** (Optional: Allow exams only from specific IP addresses/ranges)
  - **Secure Data Storage:** (This is handled by the system, no direct input from you.)
  - **Encryption Settings:** (Generally configured at the system level by technical staff, not daily admin input.)

8. Custom Grading: Flexible grading schemes and rubric management

- **What you'll input:**
  - **Grading Scheme Name:** (e.g., "Standard Percentage," "Pass/Fail," "Letter Grades A-F")
  - **Define Grade Ranges:** (e.g., "90-100% = A," "80-89% = B," "Below 50% = Fail")
  - **Add Rubric for Essay/Short Answer Questions:**
    - **Criterion Name:** (e.g., "Content Accuracy," "Clarity," "Grammar")
    - **Rating Levels:** (e.g., "Excellent," "Good," "Needs Improvement")
    - **Points per Level:** (Points assigned for each rating level)
  - **Weighting of Question Types/Sections:** (e.g., "MCQ section is 60% of total grade, Essay is 40%")

6. Financial Management & Payment Systems

Technologies: Payment gateway integration, financial analytics, accounting systems

This section handles all money-related operations, from how payments are accepted to tracking earnings and ensuring everything is financially compliant.

Payment Gateway Management: Integrate and manage multiple payment providers

What it does: Allows the platform to connect with different services (like Stripe, PayPal, Razorpay) that process payments.

Necessary Input Fields:

- **Payment Gateway Name:** (Text Field) e.g., "Stripe", "PayPal", "Razorpay"
- **API Key:** (Text Field - Encrypted/Hidden) The unique secret code provided by the payment gateway for secure connection.
- **Secret Key:** (Text Field - Encrypted/Hidden) Another secret code, often used alongside the API Key for authentication.
- **Webhook Secret:** (Text Field - Encrypted/Hidden) A secret used to verify incoming notifications from the payment gateway.
- **Environment:** (Dropdown/Radio Button) "Sandbox/Test" (for testing) or "Live/Production" (for real payments).
- **Currency Support:** (Multi-select Dropdown/Checkboxes) e.g., "INR", "USD", "EUR" (which currencies this gateway can process).
- **Default Gateway:** (Checkbox) Whether this gateway should be the primary one used for payments.
- **Status:** (Dropdown/Toggle) "Active" or "Inactive" (to enable/disable the gateway).

• Transaction Monitoring: Real-time transaction tracking and fraud detection

What it does: Keeps an eye on every payment happening on the platform as it occurs, and helps identify suspicious activities.

Necessary Input Fields (for configuration, not for each transaction):

- **Fraud Detection Rules:** (Text Area/Rule Builder Interface)
  - **Rule Name:** (Text Field) e.g., "High Value Transaction Alert"
  - **Condition:** (Dropdown + Value Input) e.g., "Transaction Amount" + "Greater Than" + "$1000"
  - **Action:** (Dropdown) e.g., "Flag for Review", "Hold Transaction", "Send Alert to Admin"
  - **Description:** (Text Area) Explains what the rule does.
- **Alert Recipients:** (Multi-select Email/User List) Who should receive notifications about suspicious transactions.
- **Alert Thresholds:** (Number Input) e.g., "Number of failed attempts before lockout", "Number of transactions from same IP in X minutes".

• Revenue Analytics: Comprehensive revenue analysis and forecasting

What it does: Provides insights into how much money the platform is making, where it's coming from, and helps predict future earnings.

Necessary Input Fields (for report generation/filtering, not for data entry):

- **Date Range:** (Date Picker - Start Date, End Date) To specify the period for analysis.
- **Revenue Type Filter:** (Multi-select Dropdown/Checkboxes) e.g., "Course Sales", "Subscription Fees", "Consulting Services".
- **Currency Filter:** (Dropdown) To view revenue in a specific currency.
- **Dimension Filters:** (Dropdowns) e.g., "By Course", "By Instructor", "By User Location".
- **Reporting Frequency:** (Dropdown) "Daily", "Weekly", "Monthly", "Quarterly", "Annually".

• Financial Reporting: Automated financial reports and audit trails

What it does: Generates various financial reports automatically (like income statements, balance sheets) and keeps a detailed record of all financial activities for auditing.

Necessary Input Fields (for report generation/configuration):

- **Report Type:** (Dropdown) "Income Statement", "Balance Sheet", "Cash Flow Statement", "Transaction History", "Audit Log".
- **Reporting Period:** (Dropdown/Date Picker) "Current Month", "Last Quarter", "Custom Range".
- **Format:** (Dropdown) "PDF", "CSV", "Excel".
- **Recipient Emails:** (Multi-select Email List) Who should receive automated reports.
- **Scheduling:** (Date/Time Picker + Frequency) e.g., "Every 1st of the Month at 9 AM".

• Refund Management: Handle refund requests and processing

What it does: Manages the process of giving money back to users for various reasons.

Necessary Input Fields:

- **Transaction ID:** (Text Field/Searchable Dropdown) To identify the original payment.
- **Refund Amount:** (Number Input) The amount to be refunded (can be full or partial).
- **Refund Reason:** (Dropdown/Text Area) e.g., "Course Cancellation", "Unsatisfactory Service", "Technical Issue", "Other (specify)".
- **Refund Type:** (Radio Button) "Full Refund" or "Partial Refund".
- **Payment Method for Refund:** (Dropdown - usually linked to original payment method) e.g., "Credit Card", "Bank Transfer".
- **Admin Notes:** (Text Area) Internal notes about the refund.

• Commission Tracking: Track and manage commission payments

What it does: Calculates and keeps records of payments due to instructors or partners based on sales or services.

Necessary Input Fields (for setting up commission rules):

- **Commission Scheme Name:** (Text Field) e.g., "Instructor Standard Commission", "Affiliate Program Tier 1".
- **Applicable To:** (Dropdown/Multi-select) "Instructor", "Affiliate", "Partner".
- **Calculation Method:** (Dropdown) "Percentage of Sale", "Fixed Amount Per Sale", "Tiered Percentage".
- **Commission Rate/Amount:** (Number Input) e.g., "10%" or "$50".
- **Trigger Event:** (Dropdown) e.g., "Course Completion", "Course Purchase", "Subscription Signup".
- **Payout Frequency:** (Dropdown) "Monthly", "Quarterly", "Upon Request".
- **Minimum Payout Threshold:** (Number Input) e.g., "$100" (minimum amount before a payout is initiated).
- **Associated User/Entity:** (Searchable User/Entity Selector) To link commission rules to specific instructors/partners.

• Tax Management: Handle tax calculations and compliance

What it does: Manages how taxes are calculated, collected, and reported, ensuring compliance with local tax laws.

Necessary Input Fields:

- **Tax Region/Country:** (Dropdown) e.g., "India", "USA", " "European Union".
- **Tax Name:** (Text Field) e.g., "GST", "VAT", "Sales Tax".
- **Tax Rate:** (Number Input - Percentage) e.g., "18%", "5%".
- **Applicability Rules:** (Rule Builder/Dropdowns) e.g., "Applies to: All Products", "Applies to: Digital Goods", "Based on User Location".
- **Tax Exemptions:** (Toggle/Checkbox) Option to mark certain products/services or users as tax-exempt.
- **Tax Reporting Frequency:** (Dropdown) "Monthly", "Quarterly", "Annually".
- **Tax ID/Registration Number:** (Text Field) The platform's own tax registration number.

• Billing Systems: Automated billing and invoice generation

What it does: Automatically creates bills and invoices for users, especially for subscriptions or recurring services.

Necessary Input Fields (for configuring billing rules):

- **Invoice Template:** (Text Area/Rich Text Editor) To customize the look and content of invoices.
- **Billing Cycle:** (Dropdown) "Monthly", "Quarterly", "Annually", "One-time".
- **Due Date Policy:** (Number Input) e.g., "7 days from invoice generation", "30 days".
- **Late Payment Penalties:** (Number Input/Rule Builder) e.g., "1% per day", "Fixed fee of $X".
- **Reminder Schedule:** (Number Input + Dropdown) e.g., "3 days before due date", "1 day after due date".
- **Invoice Prefix/Suffix:** (Text Field) e.g., "INV-", "-2025".
- **Auto-charge Enabled:** (Toggle/Checkbox) Whether to automatically charge customers for recurring payments.

• Financial Compliance: Ensure compliance with financial regulations

What it does: Ensures that all financial operations on the platform meet legal and regulatory requirements.

Necessary Input Fields (for documenting/configuring compliance settings):

- **Regulation Name:** (Text Field) e.g., "GDPR", "PCI DSS", "Local Tax Laws".
- **Description/Key Requirements:** (Text Area) Details of what the regulation entails for the platform.
- **Compliance Status:** (Dropdown) "Compliant", "Partially Compliant", "Non-Compliant".
- **Audit Log Retention Period:** (Number Input + Dropdown) e.g., "7 Years", "10 Years" (how long financial records must be kept).
- **Data Encryption Status:** (Toggle) Whether sensitive financial data is encrypted.
- **Regulatory Body Contact Information:** (Text Area/Contact Fields) Details for relevant financial authorities.

7. AI Module Management & Configuration

1. AI Model Configuration

- **Purpose:** To allow administrators to set up and fine-tune individual AI models used for recommendations (e.g., course recommendations, skill path suggestions).
- **Input Fields:**
    - **Model Name:** Text input (e.g., "Course Recommender V1", "Skill Path Predictor")
    - **Model Type:** Dropdown (e.g., "Collaborative Filtering", "Content-Based Filtering", "Deep Learning Model")
    - **Description:** Text area (brief explanation of the model's purpose)
    - **Training Data Source:** Dropdown/Multi-select (e.g., "User Activity Data", "Course Metadata", "Skill Assessments")

- **Configuration Parameters (Dynamic based on Model Type):**
  - *For Collaborative Filtering:* Number of neighbors (Numeric input), Similarity metric (Dropdown: "Cosine", "Jaccard")
  - *For Deep Learning Models:* Learning Rate (Numeric input), Number of Epochs (Numeric input), Activation Function (Dropdown: "ReLU", "N/A", "Sigmoid")
- **Status:** Dropdown (e.g., "Active", "Inactive", "Draft")
- **Save/Update Button**

---

2. Algorithm Management

- **Purpose:** To manage the specific algorithms used within AI models and their associated parameters. This might be for different recommendation strategies.
- **Input Fields:**
  - **Algorithm Name:** Text input (e.g., "Top N Recommendations", "Personalized Ranker")
  - **Associated AI Model:** Dropdown (links to a configured AI Model from section 1)
  - **Algorithm Type:** Dropdown (e.g., "Matrix Factorization", "Decision Tree", "Neural Network")
  - **Algorithm Parameters (Dynamic based on Algorithm Type):**
    - *For Matrix Factorization:* Latent factors (Numeric input), Regularization strength (Numeric input)
    - *For Decision Trees:* Max Depth (Numeric input), Min Samples Split (Numeric input)
  - **Priority/Weight:** Numeric input (for combining multiple algorithms)
  - **Status:** Dropdown (e.g., "Enabled", "Disabled")
  - **Save/Update Button**

---

3. Data Training

- **Purpose:** To initiate and manage the training process for AI models using the platform's data.
- **Input Fields:**
  - **Select AI Model(s) for Training:** Multi-select dropdown (lists configured AI Models)
  - **Training Data Range:** Date range picker (e.g., "Last 30 days", "All Time", Custom Start/End Date)
  - **Training Frequency:** Dropdown (e.g., "Daily", "Weekly", "Monthly", "Manual")
  - **Scheduled Training Time:** Time picker (if "Scheduled" frequency is selected)
  - **Start Training Button**
  - **Training Progress Indicator:** (Display only, e.g., "50% complete", "Training Successful", "Failed")
  - **Last Trained On:** (Display only, timestamp)

---

4. Performance Monitoring

- **Purpose:** To visualize and track the performance and accuracy of deployed AI models.
- **Input Fields (Mostly Display/Filter Controls):**
  - **Select AI Model:** Dropdown (for filtering performance data)
  - **Time Period:** Dropdown (e.g., "Last 24 hours", "Last 7 days", "Last 30 days")
  - **Metrics to Display (Multi-select Checkboxes):**
    - *For Recommendations:* Precision@K, Recall@K, Mean Average Precision (MAP), Coverage
    - *For Predictive Models:* Accuracy, F1 Score, RMSE, AUC
  - **Visualizations:** (Charts/Graphs for trends over time)
  - **Alert Thresholds:** Numeric input (e.g., "Alert if Accuracy drops below X%")
  - **Email for Alerts:** Text input
  - **View Detailed Logs Button**

---

5. A/B Testing

- **Purpose:** To compare the performance of different AI configurations or algorithms in a live environment to determine which performs best.
- **Input Fields:**
  - **Experiment Name:** Text input (e.g., "New Recommendation Algorithm Test")
  - **Control Group Configuration:** Dropdown (Select an existing AI Model/Algorithm configuration)
  - **Variant Group Configuration:** Dropdown (Select another AI Model/Algorithm configuration to test against)
  - **Traffic Split:** Slider/Numeric input (e.g., "50/50", "80/20")
  - **Target Metric:** Dropdown (e.g., "Click-Through Rate", "Conversion Rate", "Time Spent on Page")
  - **Duration:** Numeric input with dropdown (e.g., "7 Days", "2 Weeks")
  - **Start A/B Test Button**
  - **A/B Test Results:** (Display only, e.g., "Variant A outperforms Control by X% on Y metric")
  - **Declare Winner Button** (to deploy the better performing configuration)

---

6. Personalization Engine

- **Purpose:** To manage the rules and algorithms that drive personalized experiences for users (e.g., personalized dashboards, content feeds).
- **Input Fields:**
  - **Personalization Rule Name:** Text input (e.g., "Homepage Content Personalization")
  - **Target User Segment:** Dropdown/Multi-select (e.g., "New Users", "Learners with 5+ courses", "Specific Job Role")

- **Rule Type:** Dropdown (e.g., "Rule-Based", "Algorithm-Based")
- **If Rule-Based:**
  - **Conditions:** (Add/Remove button for multiple conditions)
    - *Condition Type:* Dropdown (e.g., "User Skill", "Course Completion", "Time on Platform")
    - *Operator:* Dropdown (e.g., "Equals", "Greater Than", "Contains")
    - *Value:* Text/Numeric/Dropdown input
  - **Action:** Dropdown (e.g., "Show Specific Content", "Prioritize Course Category", "Adjust Recommendation Weight")
  - **Action Value:** Text/Numeric/Dropdown input
- **If Algorithm-Based:**
  - **Select Algorithm:** Dropdown (e.g., "User Profile Matching", "Dynamic Content Ranking")
  - **Algorithm Parameters:** (Dynamic based on selected algorithm, similar to section 2)
- **Priority:** Numeric input (for rule conflicts)
- **Status:** Dropdown (e.g., "Active", "Inactive")
- **Save/Update Button**

---

7. Predictive Analytics

- **Purpose:** To configure models for predicting future events or trends (e.g., student churn, course completion rates).
- **Input Fields:**
  - **Predictive Model Name:** Text input (e.g., "Student Churn Predictor", "Course Completion Risk")
  - **Prediction Target:** Dropdown (e.g., "User Churn Probability", "Course Completion Likelihood", "Skill Gap")
  - **Input Features:** Multi-select (e.g., "Login Frequency", "Course Progress", "Assessment Scores", "Forum Activity")
  - **Model Type:** Dropdown (e.g., "Logistic Regression", "Random Forest", "Neural Network")
  - **Threshold for Alert/Action:** Numeric input (e.g., "Alert if churn probability > 0.7")
  - **Associated Action/Alert:** Dropdown (e.g., "Send Email to Advisor", "Flag User Account")
  - **Training Frequency:** Dropdown (e.g., "Weekly", "Monthly")
  - **Save/Update Button**

---

8. AI Ethics Compliance

- **Purpose:** To ensure that AI models are fair, unbiased, and compliant with ethical guidelines. This is more about monitoring, reporting, and setting policies.
- **Input Fields (Mostly Configuration & Reporting):**
  - **Bias Detection Metrics to Monitor:** Multi-select (e.g., "Demographic Parity", "Equal Opportunity", "Disparate Impact")

- **Sensitive Attributes:** Multi-select (e.g., "Gender", "Ethnicity", "Age", "Socio-economic Status" - *ensure compliance with data privacy regulations*)
- **Fairness Thresholds:** Numeric input (e.g., "Deviation from demographic parity should not exceed X%")
- **Ethical Review Policy Document Upload:** File Upload
- **Automated Bias Scan Schedule:** Dropdown (e.g., "Weekly", "Monthly")
- **Bias Report Generation:** Button (to generate reports on detected biases)
- **Corrective Actions/Mitigation Strategies:** Text area (for documenting actions taken)
- **Audit Log for AI Decisions:** (Display only, for transparency)

8. Platform Configuration & Customization

1. System Configuration: Configure platform-wide settings and parameters

- **What it means:** This is for general settings that affect the entire platform, like the platform's name, default language, or time zone.
- **Input Fields Examples:**
  - **Platform Name:** Text Input (e.g., "KickstartSkills Learning Platform")
  - **Default Language:** Dropdown Menu (e.g., "English", "Spanish", "Hindi")
  - **Default Time Zone:** Dropdown Menu (e.g., "IST - Indian Standard Time", "UTC - Coordinated Universal Time")
  - **Date Format:** Dropdown Menu (e.g., "DD/MM/YYYY", "MM/DD/YYYY")
  - **Currency Symbol:** Text Input (e.g., "₹", "$", "€")

---

2. Feature Management: Enable/disable features across the platform

- **What it means:** This allows the admin to turn specific functionalities on or off for all users or certain groups.
- **Input Fields Examples:**
  - **User Registration:** Toggle Switch (On/Off)
  - **Course Creation:** Toggle Switch (On/Off)
  - **Discussion Forums:** Toggle Switch (On/Off)
  - **Gamification (Badges/Points):** Toggle Switch (On/Off)
  - **Live Chat Support:** Toggle Switch (On/Off)
  - **Allow Course Reviews:** Toggle Switch (On/Off)

---

3. Theme Management: Manage platform themes and branding

- **What it means:** This controls the look and feel of the platform, including colors, logos, and fonts.

- **Input Fields Examples:**
  - **Primary Brand Color:** Color Picker (e.g., Hex code input like "#3498db" or a color palette selector)
  - **Secondary Brand Color:** Color Picker
  - **Logo Upload (Light Theme):** File Upload Button (for image files like PNG, SVG)
  - **Logo Upload (Dark Theme):** File Upload Button
  - **Favicon Upload:** File Upload Button (for .ico files)
  - **Font Family (Heading):** Dropdown Menu (e.g., "Arial", "Roboto", "Open Sans")
  - **Font Family (Body Text):** Dropdown Menu
  - **Custom CSS:** Text Area (for advanced users to paste CSS code)

---

4. White-label Configuration: Configure white-label solutions for institutions

- **What it means:** This is for organizations that want to use the platform under their own brand, with custom branding and potentially separate login portals.
- **Input Fields Examples (likely for *each* institution using white-label):**
  - **Institution Name:** Text Input
  - **Institution Logo:** File Upload Button
  - **Institution Primary Color:** Color Picker
  - **Custom Domain/Subdomain:** Text Input (e.g., "https://www.google.com/search?q=learning.institution-name.com")
  - **Custom Login Page Text:** Text Area
  - **Enable Custom Email Templates:** Toggle Switch

---

5. API Configuration: Manage API settings and integrations

- **What it means:** This is for connecting the platform with other software or services using APIs (Application Programming Interfaces).
- **Input Fields Examples:**
  - **API Key (General):** Text Input (often a long string of characters)
  - **API Secret (General):** Text Input (often a long string of characters, usually hidden/masked)
  - **Enable Specific Integration (e.g., Payment Gateway):** Toggle Switch
  - **Payment Gateway API Key:** Text Input
  - **Payment Gateway Secret Key:** Text Input
  - **Learning Management System (LMS) Integration URL:** Text Input
  - **Webhooks Enable/Disable:** Toggle Switch
  - **Webhook URL:** Text Input

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6. Notification Settings: Configure system-wide notification preferences

- **What it means:** This allows the admin to control what notifications are sent out (e.g., emails, in-app notifications) and when.
- **Input Fields Examples:**
  - **Enable Email Notifications:** Toggle Switch
  - **Enable In-App Notifications:** Toggle Switch
  - **Sender Email Address:** Text Input (e.g., "noreply@kickstartskills.com")
  - **New User Welcome Email (Enable/Disable):** Toggle Switch
  - **Course Completion Email (Enable/Disable):** Toggle Switch
  - **Password Reset Email (Enable/Disable):** Toggle Switch
  - **Notification Frequency (e.g., Daily Digest):** Dropdown (e.g., "Immediate", "Hourly", "Daily", "Weekly")
  - **SMS Notifications (Enable/Disable):** Toggle Switch (if applicable)

---

7. Maintenance Mode: Manage platform maintenance and downtime

- **What it means:** This allows the admin to put the platform into a special mode when updates or fixes are being applied, informing users it's temporarily unavailable.
- **Input Fields Examples:**
  - **Enable Maintenance Mode:** Toggle Switch
  - **Maintenance Message:** Text Area (e.g., "Our platform is currently undergoing scheduled maintenance. We'll be back shortly!")
  - **Estimated Downtime:** Text Input (e.g., "30 minutes", "2 hours")
  - **Allow Specific IP Addresses to Access:** Text Area (list of IP addresses, one per line)

---

8. Version Control: Manage platform versions and updates

- **What it means:** This is about displaying the current version of the platform and potentially managing updates. While direct input fields for *changing* the version are less common (updates are usually pushed by developers), admins might see information about it.
- **Input Fields/Display Examples:**
  - **Current Platform Version:** Display Only Text (e.g., "v2.5.1")
  - **Last Update Date:** Display Only Text (e.g., "July 10, 2025")
  - **Available Updates (if applicable):** Button (e.g., "Check for Updates", "Install Update") - *This would trigger a background process, not a direct input.*
  - **Rollback to Previous Version:** Button (likely with a confirmation dialog, used in emergencies) - *This would be a complex action, not just a simple input.*

9. Analytics & Business Intelligence

1. Business Intelligence Dashboard: Comprehensive BI dashboard with KPIs

- **Goal:** Allow admins to customize what KPIs are displayed and how.
- **Potential Input Fields:**
  - **KPI Selection Checklist:** A list of available Key Performance Indicators (e.g., "Total Users," "Course Completions," "Revenue," "Active Sessions") with checkboxes to select which ones to display on the dashboard.
  - **KPI Thresholds (Numerical Input):** For certain KPIs, allow admins to set warning or goal thresholds (e.g., "Alert me if active users drop below [500]").
  - **Dashboard Layout Configuration (Drag-and-Drop/Selection):** Options to arrange widgets or choose from pre-defined dashboard layouts.
  - **Date Range Selector (Dropdown/Calendar):** To view KPIs for specific periods (e.g., "Last 7 Days," "This Month," "Custom Range").

---

2. Custom Analytics: Create custom analytics and reports

- **Goal:** Enable admins to define specific data queries and report formats.
- **Potential Input Fields:**
  - **Report Name (Text Input):** A field to name the custom report (e.g., "Monthly Course Enrollment Report").
  - **Data Source Selection (Dropdown/Checklist):** Choose which data tables or modules to pull data from (e.g., "Users," "Courses," "Payments," "Quizzes").
  - **Metrics Selection (Checklist/Multi-select):** Select the specific data points to include in the report (e.g., "User ID," "Course Title," "Enrollment Date," "Completion Status").
  - **Dimensions/Filters (Dropdowns/Text Input):** To slice and dice the data (e.g., "Filter by Course Category [Dropdown]," "Filter by User Location [Text Input]").
  - **Grouping Options (Dropdown):** How to group the data (e.g., "Group by Month," "Group by Course," "Group by User Role").
  - **Chart Type Selection (Dropdown/Radio Buttons):** Choose how to visualize the data (e.g., "Bar Chart," "Line Graph," "Pie Chart," "Table").
  - **Scheduling Options (Dropdown/Date & Time Pickers):** To run reports automatically (e.g., "Daily," "Weekly," "Monthly," "Specific Date/Time").

---

3. Data Warehousing: Manage data warehouse and analytics infrastructure

- **Goal:** Provide controls for data ingestion, transformation, and storage. (This is more backend-focused, so input fields would be for configuration rather than direct data entry).
- **Potential Input Fields:**
  - **Data Source Connection Settings (Text Inputs/Dropdowns):** For connecting to external systems (e.g., "Database Host," "API Key," "Authentication Method").

    - **Data Sync Schedule (Dropdown/Date & Time Pickers):** How often data is updated in the warehouse (e.g., "Real-time," "Hourly," "Daily Sync at [Time]").
    - **Data Retention Policy (Numerical Input/Dropdown):** How long data is stored (e.g., "Retain data for [X] months/years," "Delete data older than [Date]").
    - **Data Anonymization Settings (Checkboxes/Dropdowns):** Options to anonymize sensitive data fields for privacy compliance.
    - **Storage Optimization Options (Checkboxes):** For enabling data compression or archiving.

---

4. Predictive Analytics: Forecast trends and predict user behavior

- **Goal:** Allow admins to configure predictive models and view forecasts.
- **Potential Input Fields:**
    - **Prediction Target Selection (Dropdown):** What to predict (e.g., "User Churn," "Course Completion Rate," "Future Enrollments").
    - **Prediction Horizon (Numerical Input/Dropdown):** How far into the future to predict (e.g., "Next 30 Days," "Next Quarter").
    - **Model Training Frequency (Dropdown/Date & Time Pickers):** How often the predictive model is retrained with new data.
    - **Key Factors for Prediction (Checklist/Multi-select):** Allow selection of data points to be considered by the model (e.g., "Login Frequency," "Time Spent on Platform," "Course Progress"). (More advanced: weighting of factors).
    - **Alert Thresholds for Predictions (Numerical Input):** Set alerts if a prediction falls below or above a certain value (e.g., "Alert if churn prediction for a user exceeds [X]%").

---

5. Market Intelligence: Analyze market trends and competitive landscape

- **Goal:** Enable admins to define competitor and market segments for analysis.
- **Potential Input Fields:**
    - **Competitor List (Text Input - Add/Remove):** A simple list where admins can add or remove competitor names/URLs.
    - **Market Segment Definition (Text Input/Checkboxes):** Define target market segments (e.g., "Education Sector," "Corporate Training," "K-12").
    - **Keyword/Topic Monitoring (Text Input - Add/Remove):** Add keywords or topics related to market trends to monitor external sources.
    - **Data Source Configuration for External Data (Text Inputs/Dropdowns):** If connecting to external market data APIs, similar to Data Warehousing connections (e.g., "API Key for Market Research Tool").

---

6. User Behavior Analytics: Deep dive into user behavior patterns

- **Goal:** Provide tools to segment users and analyze their interactions.
- **Potential Input Fields:**
  - **User Segment Definition (Text Input/Rule Builder):** Create custom user segments based on various criteria (e.g., "Users who completed Course X," "Users who logged in less than 3 times last month"). This might involve:
    - **Condition Builder (Dropdowns for "Property," "Operator," "Value"):** (e.g., "Course Status [is] [Completed]", "Last Login [before] [Date Picker]").
    - **Segment Name (Text Input).**
  - **Event Tracking Configuration (Checkboxes/Text Input):** Define what user actions to track (e.g., "Course Enrollment Click," "Video Play," "Quiz Attempt").
  - **Funnel Analysis Definition (Ordered List Input):** Define a sequence of events to track user progression (e.g., "Visit Course Page -> Add to Cart -> Purchase").
  - **Session Replay Settings (Toggle/Numerical Input):** Enable/disable session recording and set retention limits (if applicable).

---

7. Performance Benchmarking: Benchmark platform performance against industry standards

- **Goal:** Allow admins to input industry standards or select from pre-defined benchmarks.
- **Potential Input Fields:**
  - **Industry Selection (Dropdown):** Choose the relevant industry for benchmarking (e.g., "E-learning," "SaaS," "Online Course Platform").
  - **Benchmark Metrics Selection (Checklist):** Select specific metrics to compare (e.g., "Page Load Time," "Uptime," "User Engagement Rate").
  - **Custom Benchmark Input (Numerical Input):** Allow admins to manually input custom industry standards if not available from pre-defined lists.
  - **Competitor Performance Data Input (Numerical Input - if manual):** For confidential competitor data, admins might manually enter data points for direct comparison.

---

8. ROI Analysis: Calculate return on investment for various initiatives

- **Goal:** Enable admins to input costs and expected returns for initiatives.
- **Potential Input Fields:**
  - **Initiative Name (Text Input):** Name of the project or initiative (e.g., "New Course Launch," "Marketing Campaign X").
  - **Investment Costs (Numerical Input):** Input various cost categories (e.g., "Development Costs," "Marketing Spend," "Personnel Costs").
  - **Expected Revenue/Benefits (Numerical Input):** Input anticipated financial gains or qualitative benefits.
  - **Timeframe for ROI Calculation (Dropdown/Date Picker):** Define the period over which ROI is calculated.

    - **ROI Calculation Method Selection (Dropdown):** If multiple methods are supported (e.g., "Simple ROI," "Payback Period").
    - **Assumptions (Text Area):** Space to document any assumptions made during the ROI calculation.

10. Support & Service Management

1. Support Ticket System

This allows administrators to manage and track user support requests.

- **Create New Ticket:**
    - **User/Requester:** (Dropdown/Searchable list of existing users or text field for new user email/name)
    - **Subject:** (Text field - e.g., "Login Issue," "Course Enrollment Problem")
    - **Description:** (Text area - detailed explanation of the issue)
    - **Priority:** (Dropdown - e.g., Low, Medium, High, Urgent)
    - **Category/Department:** (Dropdown - e.g., Technical, Billing, Course Content)
    - **Assignee:** (Dropdown/Searchable list of support agents/teams)
    - **Status:** (Dropdown - e.g., Open, In Progress, Resolved, Closed, Awaiting User Reply)
    - **Attachments:** (File upload - for screenshots, error logs, etc.)
- **Update Existing Ticket:**
    - **Agent Reply/Internal Note:** (Text area - for communication with the user or internal notes)
    - **Status:** (Dropdown - to change the ticket's current state)
    - **Assignee:** (Dropdown - to reassign the ticket)
    - **Attachments:** (File upload)
    - **Time Spent (Optional):** (Numeric field - for tracking agent effort)

---

2. Knowledge Base Management

This enables the creation and organization of help articles and documentation.

- **Create New Article:**
    - **Title:** (Text field - concise title of the article)
    - **Category:** (Dropdown/Searchable list - e.g., Getting Started, Troubleshooting, Account Management)
    - **Tags/Keywords:** (Text field - comma-separated for searchability)
    - **Content:** (Rich text editor - for article body, including images, links, formatting)
    - **Status:** (Dropdown - e.g., Draft, Published, Archived)

    - **Author:** (Dropdown - automatically populated with the current admin/user)
    - **Visibility:** (Dropdown - e.g., Public, Internal Only)
- **Manage Categories:**
    - **Category Name:** (Text field)
    - **Description (Optional):** (Text area)

---

3. Live Chat Administration

This manages the live chat support functionality and agents.

- **Agent Management:**
    - **Agent Name:** (Text field)
    - **Agent Email:** (Text field - for login)
    - **Password:** (Password field)
    - **Availability Status:** (Toggle/Dropdown - e.g., Online, Offline, Away)
    - **Assigned Departments (Optional):** (Multi-select dropdown - for specific chat queues)
    - **Permissions/Roles:** (Dropdown - e.g., Agent, Supervisor)
- **Chat Settings:**
    - **Welcome Message:** (Text area)
    - **Offline Message:** (Text area)
    - **Chat Hours:** (Time picker/Dropdown for start and end times, days of week)
    - **Max Concurrent Chats per Agent:** (Numeric field)
    - **Pre-Chat Survey Questions:** (Text fields for defining questions - e.g., "What is your issue?")

---

4. FAQ Management

This allows for the creation and maintenance of frequently asked questions.

- **Create New FAQ:**
    - **Question:** (Text field - the common question)
    - **Answer:** (Rich text editor - the detailed answer)
    - **Category (Optional):** (Dropdown/Searchable list - to group similar FAQs)
    - **Order/Sequence (Optional):** (Numeric field - for manual sorting)
    - **Status:** (Dropdown - e.g., Draft, Published)

---

5. Service Level Management (SLAs)

This manages the agreements for response and resolution times.

- **Create New SLA Policy:**

    - **SLA Name:** (Text field - e.g., "Premium Support SLA," "General Inquiry SLA")
    - **Description (Optional):** (Text area)
    - **Applies To:** (Dropdown/Multi-select - e.g., Ticket Category, User Group, Priority Level)
    - **First Response Time Target:** (Numeric field + Dropdown for unit - e.g., 4 hours, 1 business day)
    - **Resolution Time Target:** (Numeric field + Dropdown for unit - e.g., 24 hours, 3 business days)
    - **Escalation Rules:** (Rules builder - e.g., "If not responded in 2 hours, escalate to Supervisor")
        - *Within Escalation Rules:*
            - **Time Threshold:** (Numeric field + Dropdown for unit)
            - **Action:** (Dropdown - e.g., Notify Agent, Notify Supervisor, Change Priority)
            - **Target (for notification):** (Dropdown of users/teams)

---

6. Customer Satisfaction (CSAT)

This tracks customer satisfaction and feedback.

- **CSAT Survey Settings:**
    - **Survey Trigger:** (Dropdown - e.g., After Ticket Resolution, After Chat)
    - **Survey Questions:** (Text area/Rich text editor - for the survey content)
    - **Rating Scale:** (Dropdown - e.g., 1-5 stars, Yes/No, Emoticons)
    - **Feedback Comment Field:** (Toggle - Enable/Disable)
    - **Email Template (for survey link):** (Rich text editor)
- **View CSAT Feedback:** (This would be more of a display/reporting area, but could have filters)
    - **Filter by Date Range:** (Date pickers)
    - **Filter by Agent/Category:** (Dropdowns)

---

7. Support Analytics

This analyzes support metrics and performance. (Primarily a display/reporting section with filters)

- **Reporting Filters:**
    - **Date Range:** (Date pickers - e.g., Last 7 days, This Month, Custom Range)
    - **Ticket Status:** (Multi-select dropdown)
    - **Agent:** (Multi-select dropdown)
    - **Category:** (Multi-select dropdown)
    - **Priority:** (Multi-select dropdown)
    - **SLA Compliance:** (Toggle/Dropdown - e.g., Met, Missed, All)

---

8. Escalation Management

This handles support escalations and critical issues. (Often integrated with SLA Management and Ticket System, but can have dedicated configuration)

- **Escalation Rule Configuration (if separate from SLA):**
  - **Rule Name:** (Text field)
  - **Trigger Condition:** (Rule builder - e.g., "Ticket Priority is Urgent," "No Response in X Hours")
    - *Within Trigger Condition:*
      - **Field:** (Dropdown - e.g., Priority, Last Agent Reply Date, Status)
      - **Operator:** (Dropdown - e.g., is, is not, greater than, less than)
      - **Value:** (Text/Numeric/Date field)
  - **Action:** (Dropdown - e.g., Notify User/Team, Change Priority, Reassign)
  - **Target (for action):** (Dropdown of users/teams or specific priority/assignee)
  - **Escalation Path (if multi-level):** (Ordered list of actions/targets)

11. System Monitoring & Maintenance

11.1 Real-time Monitoring

- **Description:** Allows administrators to view the current performance and health status of the system. This section is primarily about *displaying* data, but configuration might involve defining what metrics to monitor.
- **Necessary Information (for Configuration):**
  - **Metric Selection:**
    - **Input Field:** Checkbox/Multi-select list
    - **Purpose:** To allow selection of specific system metrics to monitor (e.g., CPU Usage, Memory Usage, Disk I/O, Network Latency, Active Users, Server Response Time).
  - **Dashboard Refresh Rate:**
    - **Input Field:** Dropdown/Number input
    - **Purpose:** To set how frequently the real-time data on the dashboard should update (e.g., Every 5 seconds, 10 seconds, 30 seconds, 1 minute).
  - **Visualization Type:**
    - **Input Field:** Radio buttons/Dropdown
    - **Purpose:** To choose how the data is displayed (e.g., Line Graph, Bar Chart, Gauge).

11.2 Automated Alerts

- **Description:** Sets up automatic notifications when specific system issues or anomalies occur.
- **Necessary Information (for Creating an Alert Rule):**
  - **Alert Name:**
    - **Input Field:** Text input
    - **Purpose:** A unique, descriptive name for the alert (e.g., "High CPU Usage Alert," "Database Connection Failure").
  - **Metric to Monitor:**
    - **Input Field:** Dropdown
    - **Purpose:** The specific system metric this alert will track (e.g., CPU Usage, Free Disk Space, Failed Logins, API Error Rate).
  - **Threshold Value:**
    - **Input Field:** Number input
    - **Purpose:** The critical value that, if crossed, triggers the alert (e.g., 90% for CPU, 10GB for Free Disk Space).
  - **Threshold Operator:**
    - **Input Field:** Dropdown
    - **Purpose:** How the threshold is evaluated (e.g., "Greater Than," "Less Than," "Equals," "Not Equals").
  - **Duration (Optional):**
    - **Input Field:** Number input + Dropdown (e.g., minutes, hours)
    - **Purpose:** How long the threshold must be exceeded before an alert is triggered (e.g., 5 minutes of CPU > 90%).
  - **Severity Level:**
    - **Input Field:** Dropdown
    - **Purpose:** To categorize the urgency of the alert (e.g., Critical, High, Medium, Low).
  - **Notification Channel:**
    - **Input Field:** Checkbox/Multi-select list
    - **Purpose:** Where the alert notification should be sent (e.g., Email, SMS, Slack, PagerDuty).
  - **Recipient Email/Phone/Webhook:**
    - **Input Field:** Text input (multiple lines or add more button)
    - **Purpose:** The specific addresses/numbers/URLs for the chosen notification channels.
  - **Enable/Disable:**
    - **Input Field:** Toggle switch
    - **Purpose:** To activate or deactivate the alert rule.

11.3 Performance Optimization

- **Description:** Provides tools or settings to improve system performance. This often involves displaying recommendations or allowing configuration of certain system parameters.
- **Necessary Information (for Configuration/Action):**
  - **Optimization Type:**

        - **Input Field:** Dropdown/Radio buttons
        - **Purpose:** To select the type of optimization to perform or configure (e.g., "Cache Settings," "Resource Allocation," "Indexing Configuration").
    - **Specific Parameter (based on type):**
        - **Input Field:** Varies (e.g., Number input for cache size, Toggle for enabling/disabling a feature).
        - **Purpose:** The exact setting to be adjusted.
    - **Schedule Optimization Run (if automated):**
        - **Input Field:** Date/Time picker, Frequency dropdown (e.g., Daily, Weekly, Monthly)
        - **Purpose:** To automate certain optimization tasks (e.g., log cleanup, re-indexing).

11.4 Database Management

- **Description:** Focuses on monitoring and optimizing database performance.
- **Necessary Information (for Configuration/Action):**
    - **Database Connection Details (if not auto-detected):**
        - **Input Field:** Text inputs
        - **Purpose:** Host, Port, Database Name, Username, Password (for connecting to external DB).
    - **Query Performance Monitoring Threshold:**
        - **Input Field:** Number input (milliseconds)
        - **Purpose:** To define what constitutes a "slow query" for alerting.
    - **Indexing Recommendation Toggle:**
        - **Input Field:** Toggle switch
        - **Purpose:** To enable or disable automatic recommendations for database indexing.
    - **Scheduled Optimization Tasks:**
        - **Input Field:** Date/Time picker, Frequency dropdown
        - **Purpose:** For tasks like re-indexing, vacuuming, or statistics updates.

11.5 Backup Management

- **Description:** Manages automated backups and recovery procedures.
- **Necessary Information (for Creating a Backup Job):**
    - **Backup Name:**
        - **Input Field:** Text input
        - **Purpose:** A unique name for the backup job.
    - **Data to Backup:**
        - **Input Field:** Checkbox/Multi-select list
        - **Purpose:** What specifically to back up (e.g., "All Data," "Database Only," "Application Files," "Specific Folders").
    - **Backup Frequency:**
        - **Input Field:** Dropdown

        - **Purpose:** How often the backup should run (e.g., Daily, Weekly, Monthly, Hourly).
    - **Backup Time:**
        - **Input Field:** Time picker
        - **Purpose:** The specific time of day the backup should start.
    - **Retention Policy:**
        - **Input Field:** Number input + Dropdown (e.g., days, weeks, months)
        - **Purpose:** How long backups should be stored before being deleted (e.g., keep 7 daily backups, 4 weekly backups).
    - **Storage Location:**
        - **Input Field:** Dropdown/Text input
        - **Purpose:** Where the backups should be stored (e.g., "Local Storage," "AWS S3 Bucket," "Google Cloud Storage," "FTP Server"). If external, further fields for credentials/paths.
    - **Notification on Success/Failure:**
        - **Input Field:** Checkbox + Text input for email/channel
        - **Purpose:** To receive alerts regarding backup job status.
    - **Restore Point Selection (for recovery):**
        - **Input Field:** Date/Time picker, List of available backups
        - **Purpose:** To select a specific backup to restore from.

11.6 Security Monitoring

- **Description:** Monitors system security and detects threats.
- **Necessary Information (for Configuration):**
    - **Log Source Configuration:**
        - **Input Field:** Dropdown, Text input
        - **Purpose:** To specify which logs to monitor (e.g., "Authentication Logs," "Application Logs," "Firewall Logs," "OS Logs").
    - **Security Event Thresholds:**
        - **Input Field:** Number input
        - **Purpose:** Number of failed login attempts before an alert, frequency of suspicious activities.
    - **Blocked IP Management:**
        - **Input Field:** Text area (list of IPs), Add/Remove buttons
        - **Purpose:** To manually add or remove IP addresses from a blacklist.
    - **Anomaly Detection Sensitivity:**
        - **Input Field:** Slider/Dropdown (e.g., Low, Medium, High)
        - **Purpose:** How sensitive the system should be to detect unusual patterns.
    - **Security Alert Channels:**
        - **Input Field:** Checkbox/Multi-select list + Text input
        - **Purpose:** Similar to general alerts, where security notifications should be sent.

11.7 Capacity Planning

- **Description:** Plans and manages system capacity and scaling. This often involves displaying current usage and projecting future needs.
- **Necessary Information (for Configuration/Forecasting):**
  - **Resource Usage History Period:**
    - **Input Field:** Dropdown (e.g., Last 7 days, 30 days, 90 days, 1 year)
    - **Purpose:** To define the data range for capacity analysis.
  - **Growth Rate Projection (Manual Override):**
    - **Input Field:** Number input (%)
    - **Purpose:** To manually input an expected growth rate if automatic prediction isn't sufficient.
  - **Scaling Triggers:**
    - **Input Field:** Metric dropdown, Threshold input, Action dropdown
    - **Purpose:** To configure rules for automatic scaling (e.g., "If CPU > 80% for 5 mins, add 1 server instance").
  - **Scaling Limits (Min/Max Instances):**
    - **Input Field:** Number inputs
    - **Purpose:** To set boundaries for automatic scaling to prevent over-provisioning or under-provisioning.

11.8 Maintenance Scheduling

- **Description:** Schedules and manages system maintenance tasks.
- **Necessary Information (for Creating a Maintenance Window):**
  - **Maintenance Name:**
    - **Input Field:** Text input
    - **Purpose:** A descriptive name for the maintenance activity (e.g., "Monthly Server Updates," "Database Optimization").
  - **Maintenance Type:**
    - **Input Field:** Dropdown
    - **Purpose:** What kind of maintenance (e.g., "Server Updates," "Database Schema Migration," "Application Deployment," "System Reboot").
  - **Start Date & Time:**
    - **Input Field:** Date picker, Time picker
    - **Purpose:** When the maintenance window begins.
  - **End Date & Time (Optional):**
    - **Input Field:** Date picker, Time picker
    - **Purpose:** When the maintenance window is expected to end.
  - **Recurrence:**
    - **Input Field:** Dropdown (e.g., "One-time," "Daily," "Weekly," "Monthly," "Custom")
    - **Purpose:** If the maintenance is a recurring event. If custom, further frequency fields.
  - **Affected Systems/Services:**
    - **Input Field:** Multi-select list

        - **Purpose:** Which parts of the system will be impacted by this maintenance.
    - **Downtime Expected:**
        - **Input Field:** Toggle switch, Text input (e.g., "Yes/No," estimated duration)
        - **Purpose:** To indicate if downtime is expected and how long.
    - **Communication Channels:**
        - **Input Field:** Checkbox/Multi-select list + Text input
        - **Purpose:** Where internal or external stakeholders should be notified about the upcoming maintenance.
    - **Status (e.g., Scheduled, In Progress, Completed, Canceled):**
        - **Input Field:** Dropdown (for updating status)
        - **Purpose:** To track the state of the maintenance event.

12. Integration Management & API Administration

1. Input Fields for Admin Dashboard

This section outlines potential input fields and configurations an administrator would use to manage API and integration settings within the KickstartSkills platform.

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1. API Gateway Management: Manage API gateways and endpoints

- **Purpose:** To define and control the entry points for your APIs.
- **Input Fields:**
    - **Gateway Name:** (Text Input) e.g., "Public API Gateway," "Internal Service Gateway"
    - **Base URL/Endpoint:** (Text Input) e.g., `https://api.kickstartskills.com/v1`, `http://internal-service.local/api`
    - **Description:** (Text Area) Brief explanation of the gateway's purpose.
    - **Status:** (Dropdown/Toggle) "Active," "Inactive"
    - **Associated Services/Microservices:** (Multi-select/Tags) Link to internal services exposed via this gateway.

---

2. Third-party Integrations: Configure and manage external integrations

- **Purpose:** To connect KickstartSkills with external services (e.g., payment gateways, CRM, analytics tools).
- **Input Fields (for *each* integration):**
    - **Integration Name:** (Text Input) e.g., "Stripe Payments," "Mailchimp Newsletter," "Google Analytics"

    - **Integration Type:** (Dropdown) e.g., "Payment Gateway," "Email Marketing," "Analytics," "CRM"
    - **API Key/Secret:** (Text Input - often masked) Credentials provided by the third-party service.
    - **Endpoint URL:** (Text Input) The base URL for the third-party's API.
    - **Authentication Method:** (Dropdown) e.g., "API Key," "OAuth 2.0," "Basic Auth" (with fields for client ID, client secret, redirect URI if OAuth)
    - **Configuration Parameters:** (Dynamic fields based on integration type) JSON or key-value pairs for specific settings.
    - **Status:** (Dropdown/Toggle) "Enabled," "Disabled"
    - **Last Sync/Connection Status:** (Read-only display) Indicates the last successful connection or error.

---

3. Webhook Management: Set up and manage webhooks for real-time updates

- **Purpose:** To send real-time notifications to external systems when specific events occur within KickstartSkills.
- **Input Fields (for *each* webhook):**
    - **Webhook Name:** (Text Input) e.g., "New User Registration Notification," "Course Completion Alert"
    - **Event Trigger:** (Dropdown/Multi-select) e.g., "User Created," "Course Enrolled," "Payment Successful," "Quiz Completed"
    - **Target URL:** (Text Input) The external URL to which the webhook data will be sent.
    - **Secret Key (for signing):** (Text Input - often masked) Used to verify the authenticity of the webhook.
    - **Payload Format:** (Dropdown) e.g., "JSON," "XML"
    - **Headers:** (Key-Value Pair Input) Custom HTTP headers to include.
    - **Status:** (Dropdown/Toggle) "Active," "Inactive"
    - **Last Delivery Status:** (Read-only display) Shows the success/failure of the last webhook delivery.

---

4. Integration Monitoring: Monitor integration performance and errors

- **Purpose:** To provide visibility into the health and performance of all integrations.
- **Input Fields (mostly display, but some configuration):**
    - **Alert Thresholds:** (Number Input) e.g., "Error Rate % for alerts," "Latency (ms) for alerts"
    - **Notification Recipients:** (Multi-select/Text Input) Email addresses or Slack channels to notify on errors.
    - **Log Retention Policy:** (Dropdown) e.g., "7 Days," "30 Days," "90 Days"
    - **Dashboard Filters:** (Date Range, Integration Name, Status, Error Type) for viewing logs and metrics.
    - **(Display only):**
        - **Integration Status Dashboard:** Visual representation (green/red indicators) for each integration.

        - **Error Logs:** Table displaying errors with timestamp, integration, error message, status code.
        - **Performance Metrics:** Graphs/charts for latency, request volume, success/error rates.

---

5. API Security: Manage API security and access controls

- **Purpose:** To control who can access your APIs and what they can do.
- **Input Fields:**
    - **Authentication Method:** (Dropdown) e.g., "API Key," "OAuth 2.0," "JWT"
    - **API Key Management:**
        - **Generate New API Key:** (Button)
        - **API Key List:** (Table with each key)
            - **Key ID:** (Read-only)
            - **Description:** (Text Input)
            - **Permissions/Scopes:** (Multi-select) e.g., "read:users," "write:courses," "admin:all"
            - **Expiration Date:** (Date Picker - Optional)
            - **Status:** (Dropdown/Toggle) "Active," "Revoked"
            - **Revoke Button:** (Action)
    - **Role-Based Access Control (RBAC) for API Endpoints:**
        - **Endpoint/Resource:** (Dropdown/Text Input) e.g., `/users`, `/courses/{id}`, `/payments`
        - **Required Role/Permission:** (Multi-select) Link specific API endpoints to required user roles/permissions.
    - **CORS Configuration:**
        - **Allowed Origins:** (Text Area - Comma Separated) e.g., `https://myfrontend.com`, `http://localhost:3000`
        - **Allowed Methods:** (Multi-select) e.g., "GET," "POST," "PUT," "DELETE"
        - **Allowed Headers:** (Text Area - Comma Separated)

---

6. Rate Limiting: Configure API rate limits and throttling

- **Purpose:** To prevent abuse and ensure fair usage of your APIs by limiting the number of requests clients can make.
- **Input Fields (for *each* endpoint or global):**
    - **Scope:** (Dropdown) "Global," "Per API Key," "Per User," "Per IP Address," "Per Endpoint"
    - **Endpoint/Method:** (Dropdown/Text Input - if per-endpoint) e.g., `/api/v1/users` (GET)
    - **Request Limit:** (Number Input) e.g., 100
    - **Time Unit:** (Dropdown) e.g., "Per Minute," "Per Hour," "Per Day"
    - **Burst Limit:** (Number Input - optional) Allows for temporary spikes above the sustained rate.

    - **Exclusion List:** (Text Area - Comma Separated) IPs or API keys to exempt from rate limits.
    - **Response on Exceeding Limit:** (Dropdown) e.g., "Return 429 Too Many Requests," "Queue Request"

---

7. Documentation Management: Maintain API documentation and guides

- **Purpose:** To provide clear and up-to-date information for developers using KickstartSkills APIs.
- **Input Fields (for *each* documentation piece):**
    - **Documentation Title:** (Text Input) e.g., "User Management API," "Payment Integration Guide"
    - **Content Editor:** (Rich Text Editor/Markdown Editor) For the actual API documentation (endpoints, parameters, examples).
    - **Associated API Version:** (Dropdown) Link to specific API versions.
    - **Status:** (Dropdown) "Draft," "Published," "Archived"
    - **Visibility:** (Dropdown) "Public," "Internal Only," "Private (requires authentication)"
    - **Last Updated By:** (Read-only)
    - **Last Updated On:** (Read-only)

---

8. Version Management: Manage API versions and backward compatibility

- **Purpose:** To allow for updates and changes to your APIs without breaking existing integrations.
- **Input Fields (for *each* API version):**
    - **Version Number:** (Text Input) e.g., "v1," "v2.1"
    - **Status:** (Dropdown) "Active," "Deprecated," "Retired"
    - **Release Date:** (Date Picker)
    - **Deprecation Date:** (Date Picker - if applicable)
    - **Retirement Date:** (Date Picker - if applicable)
    - **Description of Changes:** (Text Area) Markdown editor for detailing changes, new features, breaking changes.
    - **Migration Guide Link:** (Text Input - URL) Link to a guide for migrating from previous versions.
    - **Default Version:** (Toggle/Checkbox) Indicate the currently recommended version.

13. Data Management & Privacy

Technologies Utilized:

- **Data Governance Tools:** Systems for defining and enforcing data policies.

- **Privacy Management Systems:** Tools specifically designed to help meet privacy regulations.

Goal: To give administrators tools to control how data is handled, ensuring it's accurate, secure, and compliant with privacy laws.

A. Data Governance

- **Concept:** Setting the rules for how data is used, stored, and protected within KickstartSkills.
- **Admin Input Fields:**
  - **Policy Name:** A short, descriptive name for the data policy (e.g., "User Data Handling Policy," "Deletion Policy for Inactive Accounts").
  - **Policy Document Upload:** A button to upload a document (PDF, Word) containing the full policy details.
  - **Policy Status:** A dropdown (e.g., "Active," "Draft," "Under Review") to indicate the policy's current state.
  - **Last Reviewed Date:** A date picker to track when the policy was last updated or reviewed.
  - **Responsible Department/Person:** A text field or dropdown to assign ownership for the policy.

B. Privacy Compliance (GDPR, CCPA, etc.)

- **Concept:** Ensuring KickstartSkills follows all necessary legal rules for protecting user data, regardless of where users are located.
- **Admin Input Fields:**
  - **Compliance Checklist/Status:** A series of checkboxes or toggles for key compliance areas (e.g., "GDPR Compliant," "CCPA Compliant," "Data Protection Impact Assessments Conducted").
  - **Regulatory Links/Notes:** A text area to add links to specific regulations or internal notes about compliance efforts.
  - **DPO Contact Information:** Input fields for Data Protection Officer (DPO) name, email, and contact number. (If applicable)

C. Data Retention

- **Concept:** Deciding how long different types of data are kept before being deleted. This prevents storing unnecessary data indefinitely.
- **Admin Input Fields:**
  - **Data Type:** A dropdown or list of common data types (e.g., "User Profile Data," "Course Completion Records," "Payment History," "Login Logs").
  - **Retention Period:** A numerical input field with a unit dropdown (e.g., "3" years, "90" days, "Indefinite").
  - **Reason for Retention:** A text area to explain why data is kept for that specific period (e.g., "Legal Requirement," "Business Analytics," "Customer Support").
  - **Automated Deletion Schedule:** A toggle or checkbox to enable automatic deletion based on the retention policy.

D. Data Anonymization / Pseudonymization

- **Concept:** Making data anonymous (impossible to identify a person) or pseudonymous (difficult to identify without additional information) for analysis or testing, protecting privacy.
- **Admin Input Fields:**
  - **Anonymization Rules Configuration:**
    - **Select Data Fields:** A multi-select list of available data fields (e.g., "Email Address," "Full Name," "IP Address").
    - **Anonymization Method:** A dropdown for each selected field (e.g., "Hash," "Mask," "Shuffle," "Replace with Dummy Data").
    - **Trigger Anonymization Process:** A button to initiate the anonymization for selected datasets (e.g., "Run Anonymization on Test Data").

E. Consent Management

- **Concept:** Tracking and managing what users have agreed to regarding the use of their data (e.g., marketing emails, sharing data with partners).
- **Admin Input Fields:**
  - **Consent Type:** A text field or dropdown for types of consent (e.g., "Marketing Emails," "Personalized Recommendations," "Third-Party Data Sharing").
  - **Consent Statement Text:** A large text area to define the exact wording of the consent statement presented to users.
  - **Default Consent Status:** A toggle (e.g., "Opt-in by default," "Opt-out by default").
  - **Consent History Viewer:** A searchable table displaying individual user consent choices and when they were made/changed.

F. Data Export/Import

- **Concept:** Allowing administrators to get data out of the system (e.g., for reporting, data migration) and put data into the system.
- **Admin Input Fields:**
  - **Export Data Type:** A dropdown (e.g., "All User Data," "Course Enrollment Data," "Quiz Results").
  - **Date Range (for Export):** Start and end date pickers.
  - **Export Format:** A dropdown (e.g., "CSV," "JSON," "Excel").
  - **Export Button:** A button to trigger the data export.
  - **Import Data Type:** A dropdown (e.g., "New User Accounts," "Course Updates").
  - **Upload Import File:** A file upload button (e.g., accepts CSV, JSON).
  - **Import Preview/Validation:** A section to show a preview of the imported data and flag any errors before final import.
  - **Import Button:** A button to finalize the data import.

G. Audit Trails

- **Concept:** Keeping a detailed record of who accessed what data, when, and what changes were made. This is crucial for security and accountability.
- **Admin Input Fields (primarily for viewing and filtering):**
  - **User/Admin Filter:** A search bar or dropdown to filter audit logs by specific user or administrator.
  - **Action Type Filter:** A dropdown (e.g., "Data Accessed," "Data Modified," "Data Deleted," "Permission Change").
  - **Date/Time Range Filter:** Start and end date/time pickers.
  - **Affected Data Field/Record Filter:** A text field to search for specific data records.
  - **Audit Log Viewer:** A table displaying:
    - Timestamp
    - User/Admin ID
    - Action Performed
    - Affected Data (e.g., "User ID 123," "Course ABC")
    - Old Value (if modified)
    - New Value (if modified)
    - IP Address

H. Data Quality

- **Concept:** Ensuring the data in KickstartSkills is accurate, complete, and consistent. Bad data leads to bad decisions.
- **Admin Input Fields (primarily for monitoring and configuration of rules):**
  - **Data Quality Rule Configuration:**
    - **Rule Name:** A descriptive name (e.g., "Email Format Validation," "Required Fields for User Profile").
    - **Data Field Affected:** A dropdown of data fields.
    - **Validation Type:** A dropdown (e.g., "Regex Pattern," "Required Field," "Min/Max Length," "Unique Value").
    - **Error Message:** A text field for the message displayed if the rule is violated.
  - **Data Quality Dashboard:** A visual representation (charts/graphs) showing:
    - Percentage of complete records.
    - Number of validation errors per data type.
    - Trends in data quality over time.
  - **Data Cleaning Tools:** Buttons or links to tools that help identify and fix data quality issues (e.g., "Find Duplicate User Accounts," "Identify Missing Email Addresses").

14. Communication & Notification Management

1. Notification Engine: Manage Platform-wide Notifications and Alerts

What it does: This is your central hub for sending out general messages and alerts across the entire platform. Think of it as a system that pushes out important updates or warnings to all users or specific groups.

Key Input Fields:

- **Notification Title/Subject:** (Text Input) What's the main point of your notification?
- **Notification Message/Content:** (Text Area) The detailed message you want to convey.
- **Notification Type/Category:** (Dropdown/Radio Buttons) Is it an alert, an information message, a warning, or a success message? This helps users quickly understand the urgency or nature.
- **Target Audience:** (Dropdown/Checkbox List) Who should receive this notification? All users, specific roles (e.g., Students, Instructors, Admins), or custom groups?
- **Start Date/Time:** (Date/Time Picker) When should this notification become active?
- **End Date/Time (Optional):** (Date/Time Picker) If the notification should expire, when should it disappear?
- **Display Location/Placement (Optional):** (Checkbox/Dropdown) Where should the notification appear? (e.g., Top banner, In-app pop-up, Notification Bell icon).
- **Action Link/URL (Optional):** (Text Input) If you want users to click for more information, provide a link.

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2. Email Campaign Management: Create and Manage Email Marketing Campaigns

What it does: This feature allows you to design, send, and track mass emails for marketing, announcements, or specific communications to your user base.

Key Input Fields:

- **Campaign Name:** (Text Input) A name for your internal reference (e.g., "August Newsletter," "New Course Promotion").
- **Sender Name (From Name):** (Text Input) The name that appears as the sender in the recipient's inbox (e.g., "KickstartSkills Team").
- **Sender Email (From Email):** (Email Input) The email address the campaign is sent from (e.g., `info@kickstartskills.com`).
- **Reply-To Email (Optional):** (Email Input) If replies should go to a different address.
- **Subject Line:** (Text Input) The catchy title of your email that recipients see in their inbox.
- **Email Content/Body:** (Rich Text Editor) This is where you design your email. It should support text, images, links, and possibly simple HTML.
- **Target Audience/Recipient List:** (Dropdown/Checkbox List/File Upload) Who will receive this email? All users, specific user segments, or a custom list you upload.
- **Scheduling Options:** (Date/Time Picker, Radio Buttons) When should the email be sent? Immediately, or at a scheduled date/time?
- **Tracking Options (Optional):** (Checkboxes) Do you want to track opens, clicks, or bounces? (Usually default, but good to confirm).

- **Unsubscribe Link (Automatic):** (System Generated) Crucial for compliance; automatically added.

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3. SMS Integration: Integrate SMS Services for Critical Notifications

What it does: This enables you to send short text messages (SMS) directly to users' mobile phones, typically for urgent or time-sensitive alerts like password resets, verification codes, or critical announcements.

Key Input Fields:

- **SMS Content/Message:** (Text Area) The actual text message to be sent (often with a character limit).
- **Recipient Phone Number(s):** (Text Input/User Selection) The mobile number(s) to send the SMS to. This could be manually entered for one-offs, or selected from user profiles for bulk sends.
- **Sender ID/Name (Optional, Region Dependent):** (Text Input) Some countries allow a custom sender ID
- **SMS Type (Optional):** (Dropdown) Is it a transactional SMS (e.g., OTP) or a promotional SMS? This can affect delivery and regulations.
- **Scheduling Options (Optional):** (Date/Time Picker) Send immediately or schedule for later.

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4. Push Notification Management: Manage Mobile and Web Push Notifications

What it does: Push notifications are messages that pop up on a user's mobile device (even when they're not using your app) or web browser. They're great for quick updates, reminders, or encouraging re-engagement.

Key Input Fields:

- **Notification Title:** (Text Input) The main headline for the push notification.
- **Notification Message:** (Text Area) The short message content.
- **Target Audience:** (Dropdown/Checkbox List) Who should receive this push notification? All users, specific user groups, or users of a particular device type (e.g., Android, iOS, Web).
- **Action URL/Deeplink (Optional):** (Text Input) Where should the user go when they tap on the notification? (e.g., a specific course page, app section).
- **Icon/Image (Optional):** (File Upload) A small image to go with the notification.
- **Sound (Optional):** (Checkbox/Dropdown) Should the notification play a sound?
- **Scheduling Options:** (Date/Time Picker, Radio Buttons) Send immediately or schedule for later.

---

5. Communication Templates: Create and Manage Communication Templates

What it does: This allows you to pre-design layouts and content for your emails, SMS, or push notifications. This saves time, ensures consistency, and allows for dynamic content (like inserting a user's name).

Key Input Fields:

- **Template Name:** (Text Input) A clear name for your template (e.g., "Welcome Email," "Password Reset SMS," "Course Completion Push").
- **Communication Type:** (Dropdown) What kind of communication is this template for? (Email, SMS, Push Notification).
- **Template Content:** (Rich Text Editor for Email, Text Area for SMS/Push) The actual design and text of the template.
- **Subject Line (for Email/Push Templates):** (Text Input) The default subject line for emails or push notifications using this template.
- **Default Sender (Optional for Email Templates):** (Text Input) Pre-fill the sender name/email.
- **Preview Function:** (Button/Display Area) To see how the template will look before saving.

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6. Announcement System: Broadcast Important Announcements Platform-wide

What it does: Similar to the Notification Engine, but specifically for broader, more formal announcements that might appear in a dedicated section of the platform (like a news feed or dedicated announcements page).

Key Input Fields:

- **Announcement Title:** (Text Input) The headline of the announcement.
- **Announcement Content:** (Rich Text Editor) The full body of the announcement, potentially including images or links.
- **Publish Date/Time:** (Date/Time Picker) When the announcement should become visible.
- **Expiry Date/Time (Optional):** (Date/Time Picker) If the announcement should automatically be removed.
- **Visibility/Target Audience:** (Dropdown/Checkbox List) Who can see this announcement? (e.g., All users, Logged-in users, Specific roles).
- **Priority/Highlight (Optional):** (Dropdown/Checkbox) Should this announcement be highlighted or pinned to the top?

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7. Communication Analytics: Track Communication Effectiveness and Engagement

What it does: This isn't about sending messages, but about understanding how well your messages are performing. It shows you data like how many emails were opened, how many SMS messages were delivered, or how many push notifications were clicked.

Key Input Fields (for *viewing* data, not creating communications):

- **Date Range Selector:** (Date Pickers) To select the period for which you want to see analytics (e.g., Last 7 days, This Month, Custom Range).
- **Communication Type Filter:** (Dropdown) To filter by Email, SMS, Push Notification, or System Notification.
- **Campaign/Template Filter:** (Dropdown) To view analytics for a specific email campaign or template used.
- **Status Filter (Optional):** (Dropdown) Filter by "Sent," "Delivered," "Opened," "Clicked," "Bounced."
- **Export Data Button (Optional):** (Button) To download the analytics data.

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8. Multi-language Support: Manage Communications in Multiple Languages

What it does: This ensures that all your communications (templates, notifications, announcements) can be created and managed in different languages, allowing you to cater to a global audience.

Key Input Fields (These would be integrated into the fields above):

- **Language Selector/Toggle:** (Dropdown/Tabs) When creating any communication (email, SMS, notification, template), you would select or switch to the language you are working on.
- **Translated Content Fields:** (Dependent on Language Selector) For each language selected, there would be a separate input field for the content in that specific language.
- **Default Language Setting (Platform-wide):** (Dropdown in System Settings) To set the primary language for the platform and communications.

15. Compliance & Audit Management

1. Regulatory Compliance

This ensures the platform follows relevant education and business regulations.

- **Regulation Selector:** A **dropdown menu** or **checkboxes** to select the specific regulations to comply with (e.g., "GDPR," "HIPAA," "FERPA," "PCI DSS").
- **Compliance Checklist/Status:** For each selected regulation, a list of requirements with **checkboxes** or **status dropdowns** (e.g., "Compliant," "Non-Compliant," "Under Review").
- **Evidence Upload:** A **file upload button** next to each requirement to attach documentation proving compliance (e.g., policy documents, audit reports).

- **"Generate Compliance Report" Button:** To create a summary report of compliance status.

2. Audit Trail Management

This maintains detailed records of all activities within the system.

- **Activity Type Filter:** A **dropdown menu** or **checkboxes** to view specific types of audited actions (e.g., "Login Attempts," "Data Modifications," "User Creations," "Admin Actions").
- **User Filter:** A **searchable text box** or **dropdown menu** to filter audit logs by specific users.
- **Date/Time Range Selector:** A **date picker** to narrow down the audit trail to a specific period.
- **IP Address Filter:** A **text box** to search for activities originating from a particular IP address.
- **"Export Audit Log" Button:** To download the filtered audit trail in formats like CSV or JSON.

3. Compliance Reporting

This generates reports to demonstrate adherence to various regulations.

- **Report Template Selector:** A **dropdown menu** to choose a specific compliance report template (e.g., "GDPR Data Access Report," "Annual Security Audit Summary").
- **Reporting Period:** A **date range picker** or **dropdowns** (e.g., "Last Quarter," "Fiscal Year 2024") to define the report's timeframe.
- **Scope Selector: Checkboxes** or **dropdowns** to select which data or system components should be included in the report.
- **"Generate Report" Button:** To compile and display the compliance report.
- **"Export Report" Button:** To save the report in various formats (e.g., PDF, XML).

4. Policy Management

This involves creating and enforcing platform-wide operational and security policies.

- **Input Fields:**
  - **"Create New Policy" Button:** To open a form for defining a new policy.

        - **Policy Title:** A **text box** for the policy's name (e.g., "Data Retention Policy," "Acceptable Use Policy").
        - **Policy Content Editor:** A **rich text editor** to write the detailed policy text, including sections, rules, and examples.
        - **Effective Date:** A **date picker** for when the policy comes into effect.
        - **Affected Roles/Users: Multi-select dropdowns** or **checkboxes** to specify which users or roles the policy applies to.
    - **"Enforce Policy" / "Activate Policy" Toggle:** An **on/off switch** next to each policy to enable or disable its enforcement.
    - **"Acknowledge Policy" Button:** For users, a clickable button often presented on a login or notification screen, to confirm they've read and understood a new policy.

5. Risk Assessment

This conducts regular evaluations of potential risks and plans to reduce them.

- **Input Fields:**
    - **"Start New Risk Assessment" Button:** To initiate a new assessment process.
    - **Risk Identification: Text boxes** or **dropdowns** to identify specific risks (e.g., "Unauthorized Data Access," "System Downtime").
    - **Likelihood and Impact Rating: Dropdown menus** or **sliders** to assign numerical or descriptive values (e.g., "High," "Medium," "Low") for how likely a risk is and how severe its impact would be.
    - **Mitigation Strategy:** A **text area** to describe planned actions to reduce the risk.
    - **Owner Assignment:** A **dropdown menu** to assign responsibility for managing the risk to a specific person or team.
    - **"Mark as Mitigated" Checkbox/Button:** To update the status of a risk.

6. Certification Management

This manages the platform's official certifications and accreditations.

- **Input Fields:**
    - **"Add New Certification" Button:** To open a form for detailing a new certification.
        - **Certification Name:** A **text box** (e.g., "ISO 27001," "SOC 2 Type II").

        - **Issuing Body:** A **text box**.
        - **Date Obtained/Expires: Date pickers** for tracking validity.
        - **Certificate Upload:** A **file upload button** to attach the official certificate document.
    - **Renewal Reminders: Number input fields** or **dropdowns** to set automated reminders before a certification expires.
    - **"View Audit History" Button:** To access logs related to certification audits.

7. Legal Documentation

This manages all legal documents and agreements within the platform.

- **Input Fields:**
    - **"Upload New Legal Document" Button:** A **file upload button** for documents like Terms of Service, Privacy Policies, or partnership agreements.
    - **Document Type Selector:** A **dropdown menu** (e.g., "Terms of Service," "Privacy Policy," "NDA").
    - **Version Number:** A **text box** to track different versions of legal documents.
    - **Effective Date:** A **date picker** for the document's official start date.
    - **"Publish Document" Button:** To make a legal document publicly accessible (e.g., on the website's footer).
    - **"Manage Document History" Button:** To access past versions, often with a "Revert to" option.

8. Incident Response

This handles compliance-related incidents and investigations.

- **Input Fields:**
    - **"Report Compliance Incident" Button:** To open a form for submitting a new incident.
        - **Incident Category:** A **dropdown menu** (e.g., "Data Breach," "Policy Violation," "Regulatory Non-Compliance").
        - **Date/Time of Incident:** A **date and time picker**.
        - **Detailed Description:** A **text area** to explain the incident and its potential impact.
        - **Affected Parties: Text areas** or **searchable lists** for users, systems, or data involved.

- **Status Update:** On an incident management dashboard, a **dropdown menu** to change the incident's status (e.g., "Reported," "Investigating," "Remediated," "Closed").
- **Investigation Notes:** A **text area** for documenting findings and actions taken during the investigation.
- **"Assign Investigator" Selector:** A **dropdown menu** to assign a team member to the incident.
- **"Close Incident" Button:** To mark an incident as resolved.

16. Performance Optimization & Scalability

1. Performance Monitoring

This involves constantly watching the platform's health and speed across all its parts.

- **Metric Selection: Checkboxes** or a **multi-select dropdown** to choose which specific data points to monitor (e.g., "CPU Usage," "Memory Usage," "Database Query Time," "Network Latency," "Number of Active Users").
- **Threshold Settings:** For each selected metric, **number input fields** to define "warning" and "critical" levels. For example, "Alert if CPU > 80% for 5 minutes."
- **Time Period Selector: Dropdowns** like "Last Hour," "Last Day," "Custom Date Range" (with a **date range picker**) to view historical performance data.
- **Dashboard Configuration:** Within a monitoring tool (like Grafana), you'd find a visual editor with **drag-and-drop elements** and **configuration panels** where you use **text fields** for titles, **dropdowns** for data sources, and **number input fields** for refresh rates.
- **"Create Alert Rule" Button:** To open a form for defining custom alerts based on performance metrics.

2. Auto-scaling Configuration

This feature automatically adjusts the number of servers or resources based on demand.

- **Scaling Metric:** A **dropdown menu** to select the primary measurement that triggers scaling (e.g., "CPU Utilization," "Network In/Out," "Number of Requests per Second").
- **Scaling Thresholds: Number input fields** to define when to add more resources (scale out) or remove them (scale in). For example:

        - "Add server if CPU > 70% for 5 minutes."
        - "Remove server if CPU < 30% for 10 minutes."
    - **Min/Max Instances: Number input fields** to set the minimum and maximum number of servers/resources that the system should maintain. This prevents too few or too many resources.
    - **Cool-down Period:** A **number input field** (e.g., "300 seconds") to specify how long to wait after a scaling action before performing another.
    - **"Enable Auto-scaling" Toggle:** An **on/off switch** to activate automatic resource adjustment.

3. Load Balancing

This manages incoming traffic and distributes it efficiently across multiple servers.

- **Listener Port:** A **number input field** to specify which port the load balancer should listen on for incoming connections (e.g., "80" for HTTP, "443" for HTTPS).
- **Protocol Selector:** A **dropdown menu** (e.g., "HTTP," "HTTPS," "TCP") for the communication method.
- **Target Group Selector:** A **dropdown menu** or **searchable list** to select which group of servers the load balancer should send traffic to.
- **Health Check Path:** A **text field** to specify a web address on the servers (e.g., /health) that the load balancer will regularly check to ensure servers are working correctly.
- **Health Check Interval/Thresholds: Number input fields** to define how often to check and how many failed checks before marking a server unhealthy.
- **Load Balancing Algorithm:** A **dropdown menu** (e.g., "Round Robin," "Least Connections") to choose how traffic is distributed.
- **"Add Listener" / "Configure Rules" Buttons:** To add new traffic rules or routing paths.

4. Cache Management

This optimizes performance by temporarily storing frequently accessed data in a fast location (cache).

- **Cache TTL (Time To Live):** A **number input field** (e.g., "3600 seconds") to define how long data should remain in the cache before being refreshed.
- **Cache Key Patterns: Text areas** where developers define patterns for what data to cache (e.g., "cache all user profiles," "cache all product listings").

- **Cache Invalidation Triggers: Checkboxes** or **dropdowns** to set up rules for when cached data should be automatically removed (e.g., "Invalidate user cache when user profile is updated").
- **"Clear Cache" Button:** A clickable button to manually empty the cache (often used during troubleshooting or after major updates).
- **Cache Size/Memory Limits: Number input fields** or sliders to allocate memory to the caching system.

5. Database Optimization

This focuses on making database queries faster and more efficient.

- **Query Analysis Tool:** A specialized interface where users can **input (type)** database queries and see their performance impact (e.g., how long they take to run).
- **Index Creation: Text boxes** to input the names of tables and columns to create "indexes" on, which are like a book's index, making data retrieval much faster.
- **Connection Pool Size:** A **number input field** in application or database configuration files to set how many database connections are kept ready for reuse.
- **Timeout Settings: Number input fields** to define how long a query can run before it's canceled.
- **"Optimize Database" Button:** A button that might trigger automated database maintenance tasks.

6. CDN Management

CDN (Content Delivery Network) manages how static files (images, videos, etc.) are delivered quickly to users worldwide.

- **Origin Server URL:** A **text field** to specify the main server where the original content is stored.
- **Cache Behavior Rules: Forms** with **text fields** for URL patterns, **dropdowns** for caching duration, and **checkboxes** for compression settings.
- **Geo-restriction Settings: Checkboxes** or **multi-select lists** to allow or block content access from specific countries.
- **SSL/TLS Certificate Upload: File upload buttons** for security certificates for the CDN domain.
- **"Purge Cache" Button:** A button to instantly clear all cached content from the CDN (useful after updates).

7. Resource Allocation

This optimizes how computing resources (CPU, memory) are distributed among different parts of the system.

- **CPU/Memory Limits: Number input fields** (e.g., "2 CPU cores," "4GB Memory") to set the maximum resources a specific application or service can use.
- **CPU/Memory Requests: Number input fields** to specify the minimum resources an application needs to run.
- **Priority Levels:** A **dropdown menu** or **number input field** to assign a priority to different services, influencing which ones get resources first during high demand.
- **Node/Server Affinity: Text fields** or **dropdowns** to tell the system to run certain applications on specific types of servers.

8. Capacity Planning

This involves planning for future growth and ensuring the system can handle increased demand.

- **Growth Projections: Number input fields** to input anticipated user growth or transaction volume increases (e.g., "10% user growth per month").
- **Stress Test Parameters: Number input fields** to define how many simulated users or requests to send during a stress test (e.g., "Simulate 10,000 concurrent users").
- **Historical Data Range:** A **date range picker** to select which past data to use for analyzing trends.
- **"Generate Capacity Report" Button:** To create a report that forecasts future resource needs based on current usage and growth projections.
- **Resource Recommendation Display:** The output might recommend specific **number input fields** for new server sizes or quantities.

17. Security Administration & Threat Management

1. Security Policy Management

This feature allows administrators to define and enforce various security rules for the system.

- **"Create New Policy" Button:** To open a form for defining a new security policy.

        - **Policy Name:** A **text box** for the policy's title (e.g., "Password Complexity Policy," "Data Access Policy").
        - **Rule Type Selector:** A **dropdown menu** or **radio buttons** to choose the type of rule (e.g., "Password Requirements," "IP Restrictions," "Session Timeout").
        - **Rule Parameters:** Depending on the type, these could be:
            - **Password Complexity: Checkboxes** (e.g., "Require Uppercase," "Require Numbers"), **number input fields** (e.g., "Minimum Length: 8").
            - **IP Restrictions: Text areas** for lists of allowed/blocked IP addresses.
            - **Session Timeout:** A **number input field** (e.g., "30 minutes").
        - **Action on Violation: Dropdowns** (e.g., "Lock Account," "Log Event," "Send Alert").
    - **"Apply Policy" / "Enforce Policy" Button:** To activate the defined security policy.
    - **Policy Status Toggle:** An **on/off switch** next to each policy to quickly enable or disable it.

2. Threat Detection

This involves real-time identification and response to potential security threats.

    - **"Configure Detection Rule" Button:** To create a new rule for detecting specific threats.
        - **Rule Name:** A **text box**.
        - **Threat Type Selector:** A **dropdown menu** (e.g., "Unusual Login Activity," "Data Exfiltration," "Malware Detected").
        - **Thresholds: Number input fields** (e.g., "More than 5 failed logins in 1 minute," "Data transfer > 1GB in 5 minutes").
        - **Source Filter: Text fields** or **dropdowns** to specify where to look for the threat (e.g., "From specific IP," "From specific user").
        - **Response Action: Dropdowns** or **checkboxes** (e.g., "Generate Alert," "Block IP Automatically," "Suspend User Account").
    - **Alert Notification Settings: Email address text fields**, **SMS number input fields**, or **webhook URLs** to specify where alerts are sent.

- **"View Threat Dashboard" Filters: Date range pickers**, **status dropdowns** (e.g., "Active Threats," "Resolved Threats"), and **severity filters** (e.g., "High," "Medium")

3. Vulnerability Management

This focuses on regularly assessing the system for weaknesses and applying necessary fixes.

- **"Schedule Scan" Button:** To set up automated vulnerability scans.
  - **Scan Frequency: Dropdowns** (e.g., "Weekly," "Monthly").
  - **Scan Scope: Checkboxes** or **text areas** to specify which parts of the system to scan (e.g., "All Servers," "Specific Application," "Network Devices").
  - **Notification Recipients: Text fields** for email addresses to receive scan reports.
- **"Run Manual Scan" Button:** To initiate an immediate vulnerability scan.
- **"Mark as Patched" Checkbox/Button:** On a list of detected vulnerabilities, a **checkbox** or **button** to mark a vulnerability as resolved after applying a fix.
- **Exclusion List:** A **text area** to add specific vulnerabilities or assets that should be temporarily excluded from scans (with a **reason text field**).

4. Access Control Management

This manages who has access to what resources within the system.

- **User Selector:** A **searchable list** or **dropdown menu** to select a user or group.
- **Resource Selector:** A **searchable list** or **tree view** to select files, folders, applications, or data sets.
- **Permission Checkboxes:** A detailed list of actions (e.g., "Read," "Write," "Delete," "Execute," "Administer") with **checkboxes** to grant or deny specific permissions for the selected user/group on the selected resource.
- **"Add/Remove Access" Buttons:** To apply or revoke permissions.
- **Inheritance Settings: Checkboxes** or **radio buttons** to determine if permissions are inherited from parent folders/groups.

5. Encryption Management

This involves managing the keys and protocols used for data encryption.

- **Encryption Key Generation:** A **"Generate New Key" button** (often with **dropdowns** for key length/algorithm).
- **Key Rotation Schedule: Dropdowns** (e.g., "Every 90 days," "Annually") to automate the process of creating new encryption keys.
- **Certificate Upload: File upload buttons** for SSL/TLS certificates and their private keys.
- **Encryption Protocol Selector: Dropdowns** to choose the specific encryption protocol for data in transit or at rest (e.g., "TLS 1.3," "AES-256").
- **"Encrypt Data" / "Decrypt Data" Buttons:** To manually encrypt or decrypt specific data sets (for technical operations).

6. Security Incident Response

This handles procedures for responding to and resolving security incidents and breaches.

- **"Report Incident" Button:** To open a form for reporting a new security incident.
  - **Incident Type:** A **dropdown menu** (e.g., "Unauthorized Access," "Malware Infection," "Data Breach").
  - **Severity Level: Radio buttons** or a **dropdown** (e.g., "Critical," "High," "Medium").
  - **Description:** A **text area** for details about the incident.
  - **Affected Systems/Users: Text areas** or **searchable lists**.
  - **"Submit Incident Report" Button.**
- **Incident Status Update:** On an incident management dashboard, a **dropdown menu** to change the status (e.g., "New," "Investigating," "Resolved," "Closed").
- **Action Log:** A **text area** to record steps taken during incident response.

7. Penetration Testing

This coordinates regular simulated attacks to find system vulnerabilities.

- **"Schedule Pen Test" Button:** To set up a new penetration test.
  - **Test Type: Dropdowns** (e.g., "External Network," "Web Application," "Internal System").
  - **Scope Definition:** A **text area** to list IP addresses, URLs, or system components to be tested.

        - **Testing Firm Selector:** A **dropdown menu** to choose a pre-approved testing vendor.
        - **Start Date/End Date: Date pickers** for the testing period.
    - **Report Upload:** A **file upload button** for the penetration test report.
    - **Findings Tracking:** Input fields for each finding (e.g., **Vulnerability Name text box**, **Severity dropdown**, **Status dropdown** like "Open," "Fixed," "Accepted Risk").

8. Security Compliance

This ensures the system adheres to relevant security standards and regulations.

- **Compliance Standard Selector: Dropdowns** (e.g., "ISO 27001," "GDPR," "HIPAA").
- **Control Checklist:** For each standard, a list of controls with **checkboxes** to mark compliance status.
- **Evidence Upload:** Next to each control, a **file upload button** to attach documentation (e.g., policy documents, audit reports).
- **"Generate Compliance Report" Button:** To create a report summarizing adherence to selected standards.
- **Audit Log Review:** Filters (similar to log aggregation) to review specific security audit trails.

18. Reporting & Document Management

1. Executive Reporting

This feature allows creating high-level reports and dashboards, usually for decision-makers.

- **Report Template Selector:** A **dropdown menu** to choose a pre-defined executive report template (e.g., "Quarterly Performance Summary," "Student Enrollment Trends").
- **Date Range Selector:** A **date picker** or **dropdowns** (e.g., "Last Quarter," "This Year") to define the reporting period.
- **Data Filters: Dropdowns** or **checkboxes** to include/exclude specific data points (e.g., "Exclude inactive students," "Include only top-performing mentors").
- **"Generate Report" Button:** To compile and display the executive report.

- **"Export" Button:** To save the report in a chosen format (e.g., PDF, CSV).

2. Custom Report Builder

This provides advanced tools to build reports from various data sources.

- **"Create New Report" Button:** To start building a custom report.
- **Report Name:** A **text box** for the title of the custom report.
- **Data Source Selector:** A **dropdown menu** or **checkboxes** to choose which data tables or sections to pull information from (e.g., "Student Profiles," "Task Completions," "Feedback Ratings").
- **Field Selector:** A **multi-select list** or **checkboxes** to pick which specific data fields to include in the report (e.g., "Student Name," "Mentor Name," "Task Due Date," "Grade").
- **Filter Criteria: Text boxes**, **dropdowns**, or **date pickers** to set conditions for the data (e.g., "Tasks where status is 'Completed'," "Students with GPA > 3.0").
- **Sorting Options: Dropdowns** to choose how the data should be ordered (e.g., "Sort by Student Name A-Z," "Sort by Date Descending").
- **Aggregation Functions: Dropdowns** to apply calculations like "Sum," "Average," "Count" to numerical fields.
- **"Save Report" Button:** To save the custom report definition for future use.
- **"Run Report" Button:** To generate the report based on the defined criteria.

3. Automated Reporting

This allows scheduling reports to be generated and delivered automatically.

- **Report Selector:** A **dropdown menu** to choose an existing executive or custom report to automate.
- **Frequency: Radio buttons** or a **dropdown menu** to set how often the report should be generated (e.g., "Daily," "Weekly," "Monthly," "Quarterly").
- **Start Date/Time:** A **date and time picker** to specify when the automation should begin.
- **Recipients:** A **text area** for email addresses or a **searchable list** of system users to whom the report should be sent.
- **Delivery Format: Radio buttons** or a **dropdown menu** to select the file format for delivery (e.g., "PDF," "CSV," "Email Body").
- **"Schedule Report" Button:** To activate the automated generation and delivery.

4. Document Templates

This enables creating and managing templates for various documents.

- **"Create New Template" Button:** To open a rich text editor or document builder interface.
- **Template Name:** A **text box** for the template's title (e.g., "Student Enrollment Letter," "Certificate of Completion").
- **Document Content Editor:** A **rich text editor** (like a word processor) where users can type the main text, add formatting, and embed placeholders.
- **Placeholder Insertion:** A **dropdown menu** or **buttons** to insert dynamic data placeholders (e.g., [StudentName], [CompletionDate], [MentorSignature]) that will be filled automatically when a document is generated from the template.
- **"Upload Template" Button:** To upload an existing document (e.g., a Word file) to convert into a template.
- **"Save Template" Button:** To store the template for future use.

5. Digital Signatures

This implements capabilities for digitally signing documents.

- **"Request Signature" Button:** On a document, a button to initiate a signature request.
- **Signer Selector:** A **searchable list** or **text area** for email addresses to specify who needs to sign the document.
- **Signature Placement:** A **drag-and-drop interface** to place signature fields on the document.
- **Signing Method Selector: Radio buttons** or **dropdowns** to choose the signing method (e.g., "Draw Signature," "Type Signature," "Upload Signature Image").
- **"Sign Document" Button:** For the signer, a button to apply their digital signature.
- **"Send for Signature" Button:** To send the document to the designated signers.

6. Version Control

This manages different versions of documents and their approval workflows.

- **"Upload New Version" Button:** A **file upload button** on a document's page to replace it with a newer version.
- **Version Comments:** A **text area** to add notes about changes in the new version.
- **"Revert to Version" Button:** A button next to previous document versions to restore an older one.
- **Approval Workflow Setup (for administrators):**
  - **Approver Selector:** A **searchable list** or **dropdown** to select who needs to approve a document version.
  - **Approval Steps: Input fields** to define sequential approval steps (e.g., "Draft -> Reviewer 1 -> Reviewer 2 -> Final Approval").
  - **"Approve" / "Reject" Buttons:** For approvers, buttons to accept or decline a document version.

7. Document Security

This secures document storage and access control.

- **Access Permissions:** For each document or folder, **multi-select lists**, **checkboxes**, or **dropdowns** to assign "Read," "Write," "Delete" permissions to specific users or roles.
- **Encryption Settings: Toggles** or **checkboxes** to enable encryption for documents at rest.
- **Expiration Date:** A **date picker** to set a date after which a document automatically becomes inaccessible or is deleted.
- **Watermark Settings: Text fields** or **image upload buttons** to add watermarks to documents when viewed or printed.
- **"Share Document" Button:** To open a form with **email address text fields** and **permission dropdowns** for sharing.

8. Export Capabilities

This allows exporting reports and documents in multiple formats.

- **"Export" Button:** Present on report and document viewing pages.
- **Format Selector:** A **dropdown menu** that appears after clicking "Export," offering choices like "PDF," "CSV," "Excel (.xlsx)," "Word (.docx)," "JSON."
- **"Download" Button:** To initiate the download of the selected format.

19. Mobile Administration & Management

1. Mobile App Configuration

This involves setting up various settings and features of the mobile application.

- **Feature Toggles: On/off switches** or **checkboxes** for enabling/disabling specific mobile app features (e.g., "Enable Offline Mode," "Allow Camera Access").
- **Default Settings: Text boxes**, **dropdowns**, or **number input fields** to set default values for app behaviors (e.g., "Default Notification Sound," "Session Timeout (minutes)").
- **Branding Elements (if separate from web): File upload buttons** for mobile-specific logos or splash screens.
- **API Endpoints: Text fields** to specify the addresses the mobile app connects to for data.
- **"Save Configuration" Button:** To apply changes to the mobile app's settings.

2. Push Notification Management

This feature allows administrators to manage and send notifications to mobile users.

- **Notification Title:** A **text box** for the main heading of the push notification.
- **Notification Message:** A **text area** for the body of the message.
- **Target Audience: Dropdowns**, **checkboxes**, or a **searchable list** to select which users or groups should receive the notification (e.g., "All Students," "Mentors in X Group," "Specific User").
- **Deep Link URL:** A **text field** for a specific link within the app that the notification should open.
- **Schedule Send Time:** A **date and time picker** to send the notification immediately or at a scheduled time.
- **"Send Notification" / "Schedule Notification" Button:** To dispatch or schedule the push alert.
- **"View Sent Notifications" / "Notification History" List:** A display of past notifications, possibly with "Resend" or "Delete" buttons.

3. Mobile Analytics

This tracks mobile app usage and performance data.

- **Time Period Selector: Dropdowns** (e.g., "Last 7 Days," "This Month," "Custom Date Range") or a **date range picker** to filter analytics data.
- **Metric Selector: Checkboxes** or a **multi-select dropdown** to choose which usage metrics to display (e.g., "Daily Active Users," "Session Duration," "Crash Rate," "Feature Usage").

- **Platform Filter: Radio buttons** or **checkboxes** to view analytics for "iOS," "Android," or "All."
- **Version Filter:** A **dropdown menu** to analyze data for specific app versions.
- **"Generate Report" / "View Analytics" Button:** To update the displayed charts and data based on selections.

4. App Store Management

This involves managing the app's listing and updates on app stores (Apple App Store, Google Play Store).

- **App Name:** A **text box** for the app's official name in the store.
- **App Description:** A **text area** for the detailed description visible to users.
- **Keywords/Tags:** A **text field** or **multi-select list** for search optimization.
- **Category Selection:** A **dropdown menu** to choose the app's category.
- **Screenshots/Promotional Images: File upload buttons** for marketing visuals.
- **Release Notes:** A **text area** to describe new features and bug fixes for a new version.
- **Version Number:** A **text box** for the new app version (e.g., "1.2.0").
- **"Submit for Review" / "Publish Update" Button:** To send the new version to the app stores.

5. Mobile Security

This involves implementing policies and controls to secure the mobile app and data.

- **PIN/Biometric Enforcement: Checkboxes** or **toggles** to require PIN or biometric authentication on app launch.
- **Data Encryption Settings: Radio buttons** or **dropdowns** to configure encryption levels for data stored on the device.
- **Remote Wipe/Lock: Buttons** (often requiring confirmation via a **text input** for password) to remotely erase or lock a lost/stolen device's app data.
- **Jailbreak/Root Detection: On/off switches** to enable detection of compromised devices and define actions (e.g., "Block Access").
- **"Save Security Policies" Button:** To apply security changes.

6. Device Management

This manages access and compliance of mobile devices connected to the system.

- **Device Whitelist/Blacklist: Text areas** or **upload buttons** to manage lists of allowed or blocked device IDs.
- **Allowed OS Versions: Number input fields** or **ranges** to specify minimum and maximum operating system versions allowed to access the app.
- **"View Devices" List:** A display of connected devices, often with **"Block," "Unblock,"** or **"Wipe Data" buttons** next to each device.
- **Compliance Rules: Checkboxes** or **dropdowns** to enforce specific device security settings (e.g., "Require device passcode").

7. Offline Capability

This configures how the mobile app works when there's no internet connection.

- **"Enable Offline Mode" Toggle:** An **on/off switch** to activate offline functionality.
- **Data Sync Frequency:** A **dropdown menu** (e.g., "Every 1 Hour," "On App Launch," "Manual Sync") to set how often data is synchronized when online.
- **Downloadable Content Selector: Checkboxes** or a **multi-select list** to choose which content (e.g., "Course Materials," "Assigned Tasks") should be available offline.
- **Storage Limits:** A **number input field** to set the maximum amount of data the app can store offline.

8. Cross-platform Support

This involves managing different versions of the app (iOS, Android, web).

- **Version Number Input: Text fields** to specify the version number for each platform (e.g., "iOS Version: 1.2.0," "Android Version: 1.2.1").
- **Platform-Specific Feature Toggles: Checkboxes** or **toggles** to enable/disable features only for specific platforms.
- **Build Triggers: Buttons** to initiate builds for different platforms (e.g., "Build iOS App," "Build Android App").
- **Release Notes Editor:** A **text area** to write platform-specific release notes.

20. Technical Implementation Stack

Frontend Technologies:

This section describes the tools used to build what you **see and interact with** on a website or application.

1. **React.js 18+ with TypeScript for robust type safety:**
    - **Input Fields (for developers):** Developers **write code** in TypeScript (a version of JavaScript) using the React.js framework. This code defines how buttons, text boxes, and other elements (which *are* the input fields for end-users) behave and what data they handle. The "input" is the **source code** itself. TypeScript adds "type safety," meaning developers explicitly "input" what kind of data (e.g., text, number) a variable should hold, preventing common errors.
2. **Next.js for server-side rendering and optimization:**
    - **Input Fields (for developers/config):** Developers **write code** and **configuration files** within Next.js. This "input" tells Next.js how to build the website pages, whether to generate them on the server or in the user's browser, and how to optimize them for speed.
3. **Redux Toolkit for complex state management:**
    - **Input Fields (for developers):** Developers **write code** using Redux Toolkit to define how data flows and is stored across different parts of the application. They "input" rules for how data can be changed and accessed, especially in large and complex applications.
4. **Material-UI or Ant Design for enterprise-grade components:**
    - **Input Fields (for developers):** These are libraries that provide pre-built, polished interface elements (like buttons, forms, tables). Developers "input" by **writing code** that uses these components and customizes their properties (e.g., Button color="primary", TextField label="Username"). They might also "input" styling rules.
5. **React Router for advanced routing and navigation:**
    - **Input Fields (for developers):** Developers **write code** using React Router to define the different "pages" or "views" of the application and how users navigate between them when they click links or type URLs. The "input" tells the application which component to show for a given web address.
6. **Recharts/D3.js for sophisticated data visualization:**
    - **Input Fields (for developers):** Developers **write code** that takes data and "inputs" it into Recharts or D3.js functions. These functions then "output" complex charts and graphs. Developers also "input" parameters to customize the appearance and behavior of these visualizations (e.g., chartType="bar", xAxisLabel="Months").

7. **React Hook Form for efficient form management:**
    - **Input Fields (for developers):** This library helps developers manage the actual input fields (text boxes, dropdowns, etc.) that users interact with. Developers "input" by **writing code** that defines rules for these forms, such as:
        - **Validation Rules:** "Inputting" if a field is required, minimum/maximum length, or specific format (e.g., email: true).
        - **Default Values:** "Inputting" what should appear in a field when the form first loads.
        - **Submission Logic:** "Inputting" what happens when a user clicks the "Submit" button on a form.

Backend Integration (for Developers/Technical Configuration)

1. RESTful API Architecture with Comprehensive Endpoints

RESTful APIs are like a set of standardized rules that allow different software systems to communicate over the internet. "Endpoints" are specific addresses for accessing data.

- **Defining the API paths:** They decide what addresses exist (e.g., /students, /mentors/{id}). This is done by **writing code** that maps these paths to specific actions.
- **Specifying data formats:** They choose how data will be sent and received (e.g., **JSON** or **XML**).
- **Designing the "requests" and "responses":** They determine what information is sent in a request (e.g., student ID to get their profile) and what information is sent back in a response (e.g., the student's name, email, etc.). This is done by **coding the data structures**.

2. GraphQL for Efficient Data Querying and Manipulation

GraphQL is a more flexible way for different software parts to ask for exactly the data they need, nothing more, nothing less.

- **Writing GraphQL Schemas:** Developers "input" the structure of their data and the types of queries that can be made by **writing code** in a special schema definition language.
- **Crafting GraphQL Queries/Mutations:** The client (e.g., the website) "inputs" its data request by **writing a specific query** that asks for only the fields it needs. This query is sent to the backend.

3. Real-time WebSocket Connections for Live Updates

WebSockets enable a constant, open connection between a user's browser and the server, allowing for instant, live updates (like chat messages or real-time progress).

- **Defining WebSocket Endpoints:** Developers "input" by **writing code** to specify which specific "channels" or "topics" can be subscribed to for real-time updates (e.g., a "student progress" channel, a "new notification" channel).
- **Sending Messages:** The backend "inputs" real-time data by **sending messages** through the open WebSocket connection whenever something relevant changes.

4. Microservices Architecture for Scalability

Microservices break down a large software application into many small, independent services that work together. This makes the system easier to build, update, and scale.

- **Designing Service Boundaries:** Developers "input" by **deciding how to split** the application into smaller services (e.g., a "User Service," a "Task Service," an "Authentication Service"). This is a design decision and involves **coding each service separately**.
- **Defining Communication Protocols:** They "input" how these smaller services will talk to each other (e.g., using REST APIs, message queues).

5. Message Queuing for Asynchronous Processing

Message queuing is like a postal service for software. Instead of one part of the system waiting for another to finish a task, it sends a "message" to a queue and moves on, allowing tasks to be processed in the background.

- **Defining Message Types/Formats:** Developers "input" by **coding the structure** of the messages that will be sent through the queue.
- **Writing "Producers" and "Consumers":** They **write code** for the parts of the system that "produce" (send) messages to the queue and the parts that "consume" (read and process) messages from the queue.
- **Configuring Queue Settings:** In the message queue software itself, administrators might "input" settings like the queue name, message retention time, etc., using **configuration files** or administrative interfaces.

6. Caching Strategies with Redis for Performance

Caching stores frequently accessed data temporarily in a fast location (like Redis) so that the system doesn't have to retrieve it from slower sources (like a database) every time.

- **Deciding What to Cache:** Developers "input" by **writing code** that explicitly tells the system which data should be stored in the cache (e.g., frequently viewed mentor profiles, common settings).
- **Defining Cache Invalidation Rules:** They "input" by **coding rules** for when cached data should be removed or updated (e.g., "remove mentor profile from cache when updated").

- **Configuring Redis:** Technical users might "input" settings in Redis configuration files, like memory limits or security settings.

7. Database Optimization with Connection Pooling

Database optimization ensures that the database runs efficiently. Connection pooling reuses existing database connections instead of creating a new one every time, which speeds things up.

- **Writing Optimized Queries:** Developers "input" by **writing efficient SQL queries** to retrieve or store data in the database.
- **Configuring Connection Pool Size:** Database administrators or developers "input" **numerical values** (e.g., a configuration setting) to define how many database connections should be kept ready in the pool.
- **Indexing Strategies:** Database administrators "input" by **creating indexes** on database tables, which are like a book's index, making data retrieval faster. This is done through database commands.

Infrastructure & DevOps (for Technical Users)

1. Cloud Infrastructure (AWS/GCP/Azure) Management

This involves setting up and managing servers, databases, and networks in cloud environments.

- **Resource Creation Forms:** On a cloud provider's website, you'd find forms with **text boxes** for names, **dropdowns** for regions/sizes (e.g., "VM Size," "Database Type"), **number input fields** for capacity, and **checkboxes/toggles** for features (e.g., "Enable Auto-scaling").
- **Configuration Files (YAML/JSON):** These are text-based files where you define settings using specific syntax. Technical users "input" by **writing code** in these files.
- **Command Line Interface (CLI) Commands:** Text commands typed into a terminal to create, modify, or delete cloud resources. The "input" is the **command string** itself with various parameters.

2. Containerization with Docker and Kubernetes

This involves packaging applications into isolated "containers" (Docker) and managing them across many servers (Kubernetes).

- **Dockerfile (Code):** A text file where developers **write instructions** (input) for how to build a Docker container image.

- **Kubernetes YAML Manifests (Code):** Text files where operations teams **define** (input) how applications should run, scale, and connect in a Kubernetes cluster.
- **Docker/Kubernetes Commands:** Commands typed into a terminal to build, run, stop, or manage containers and clusters. The "input" is the **command string** and its arguments.

3. CI/CD Pipeline Management

CI/CD (Continuous Integration/Continuous Delivery) automates the process of building, testing, and deploying software.

- **Pipeline Configuration Files (YAML/Groovy):** Developers **write code** in these files to define the steps of the CI/CD pipeline (e.g., "build this code," "run these tests," "deploy to this server").
- **Trigger Settings: Dropdowns**, **checkboxes**, or **text fields** to define when a pipeline should run (e.g., "On every code commit," "Every night at 2 AM").
- **Environment Variables: Text fields** to define variables (like API keys or database connections) used during the pipeline execution.
- **"Run Pipeline" Button:** A clickable button to manually start a pipeline.

4. Infrastructure as Code (IaC) with Terraform

IaC manages infrastructure (servers, networks, etc.) using code, making it repeatable and version-controlled.

- **Terraform Configuration Files (.tf files):** These are text files where infrastructure engineers **write code** (input) to describe the desired state of their infrastructure. They define resources, their properties, and how they relate.
- **Variables: Text fields** within the .tfvars files or command-line arguments to provide specific values (e.g., "number of servers," "server region") to the IaC code.
- **Terraform Commands:** Commands like terraform init, terraform plan, terraform apply typed into a terminal to manage the infrastructure.

5. Monitoring and Alerting with Prometheus/Grafana

These tools collect data from systems and alert teams when issues arise.

- **Query Editor:** A **text area** where users **write code/queries** (e.g., Prometheus Query Language or PromQL) to define what data to collect and display.
- **Dashboard Creation: Drag-and-drop elements** and **configuration panels** with **text fields** for titles, **dropdowns** for data sources, and **number input fields** for refresh rates.
- **Alert Rule Definition: Forms** with **text fields** for alert names, **dropdowns** for alert conditions (e.g., "CPU usage > 90%"), **number input fields** for thresholds, and **multi-select dropdowns** for notification channels (e.g., "Send email to ops team").

6. Log Aggregation and Analysis

Collecting all system logs in one place for easier searching and troubleshooting.

- **Search Bar:** A **text box** to type in keywords, error messages, or specific values to search through logs.
- **Date Range Selector:** A **date picker** or **dropdowns** to specify the time period for log analysis.
- **Filter Options: Checkboxes**, **dropdowns**, or **text fields** to filter logs by source (e.g., "Web Server," "Database"), severity (e.g., "Error," "Warning"), or application.
- **"Create Alert" Button:** To create an alert based on a specific log pattern (similar to monitoring alerts).

7. Disaster Recovery and Backup Management

Setting up systems to recover from failures and manage data backups.

- **Backup Policy Configuration: Forms** with **dropdowns** for backup frequency (e.g., "Daily," "Weekly"), **date/time pickers** for schedule, **number input fields** for retention periods (e.g., "Keep for 30 days").
- **Resource Selection: Checkboxes** or **searchable lists** to select which servers, databases, or files to back up.
- **"Initiate Backup" Button:** To manually start a backup process.
- **"Restore Data" Button:** To start a data recovery process. This would then require **input fields** to select the backup version, target location, etc.